

Active noise control systems

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Indexing terms: Acoustics, Control systems, Noise and interference, Reviews of progress

Abstract: A retrospective review of the development of active noise control systems is presented, arguing that the design of active noise control (ANC) systems should be considered from a control systems point of view. This approach provides a design methodology that accounts for the design parameters of the system which determine its performance, thereby producing an ANC system that reduces the problems associated with, and the limited practical success of, previous techniques. Based on this argument, the fundamental conditions required for cancellation are derived in terms of the power spectral densities of the primary and secondary waves. These conditions are in turn related to the geometry-related (incorporating the acoustic response of the propagation medium) and source-related parameters of the system. From these conditions, the control structures employed in current ANC systems are examined and compared with the reported applications. A method for the design of controllers for use in ANC systems with broadband compact noise sources suitable for implementation on digital signal processing devices is presented. Using this method, experimental results using differing controllers are illustrated and discussed for both synthetic and practical sources. Finally, current developments in ANC systems are summarised and areas for further work are suggested.

1 Introduction

Unwanted acoustic noise has many negative effects on humans and animals. Numerous investigations [1-7] have shown that noise can have an adverse psychological effect on people; can damage hearing; can create an inefficient working environment; can affect the quality and quantity of sleep; can induce dangerous vibrations in buildings; and can generally put the health of people at risk. Unfortunately, this environmental pollution is becoming more common as our desire for more powerful machinery and quest for increased automation continues.

Two distinct methods have been used to reduce the sound intensity of unwanted noise: passive and active noise control. Passive noise control utilises the absorp-

tion property of matter, that is, sound absorbent materials are mounted on and/or around the source of noise. This technique is effective at higher frequencies, say, more than 500 Hz, but for lower frequencies the bulk, and hence the cost, of the absorbent material increases exponentially, making passive techniques impractical and uneconomical.

Active noise control (ANC) uses the intentional superposition of acoustic waves to create a destructive interference pattern such that a reduction of the unwanted sound occurs. A (primary) source of acoustic noise located in a propagation medium emits (unwanted) sound. A (secondary, or cancelling) artificially generated source is introduced which emits sound waves that are out of phase with the unwanted sound so that when superimposed on each other the two waves destructively interfere and reduction of the unwanted sound occurs. The generation of a stable destructive interference pattern is only possible for low frequencies, say less than 500 Hz, and for within limited areas. This restriction of frequency range is complementary to the limitations of passive methods. Any practical solution would require a combination of passive techniques for higher frequencies (greater than 500 Hz, say) and active techniques for low frequencies (less than 500 Hz, say).

The prospect of active noise control has intrigued, and frustrated, scientists and engineers in many countries for over 50 years. The basic idea is relatively simple, however, the limitations of the available electronic technology have restricted practical implementation to a few simple applications. With the recent development of fast digital signal processing devices, capable of processing signals within the audio-frequency range, an exciting opportunity now exists to make ANC systems a practical reality. This places the design of ANC systems firmly within the realm of electronic engineering. In particular, the systems currently being developed employ the analysis skills of control engineers and the design expertise of electronic engineers. Active noise control is, potentially, a significant area of application for control and electronic technology. Moreover, with the development of solid-state electronics making sophisticated signal processing realisable and inexpensive, a tradeoff between rigorous acoustic analysis and control system design is possible. These considerations combine to make ANC an important and relevant area for development by the electrical engineering community.

Although the first theoretical proposals on active noise control were reported as early as 1933, the area is still under development, and apart from a few special cases practically successful ANC systems have still to find robust practical applications.

Paper 5307A, received in final form 17th November 1986. Commissioned IEE Review

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2 Development of active noise control systems

Active noise control was one of the earliest applications of electronics to the control of physical systems. Lueg filed for a patent in Germany in 1933 and in the USA in 1934 and was granted US Patent 2043 416 in 1936 [8]. In his patent, Lueg made use of the two basic principles of realising an ANC system: interference and absorption. The interference principle results from the arbitrary physical mixing of acoustic waves resulting in constructive and destructive interference which, in turn, cause intensification and weakening of the sound field, respectively. He attempts to manipulate this principle of superposition so that the destructive interference of sound waves could be used to eliminate unwanted noise. He introduced the concept of active attenuation of sound by using artificially produced acoustic waves mixed with the unwanted sound so that the waves were in antiphase, and destructive interference resulted by design. Lueg also uses the absorption principle by synchronising the movements of a loudspeaker's diaphragm in antiphase to the unwanted noise so that the noise energy is absorbed by the loudspeaker. Lueg's proposals for obtaining destructive interference and sound energy absorption are shown in Fig. 1, which is taken from the illustration page of his patent.

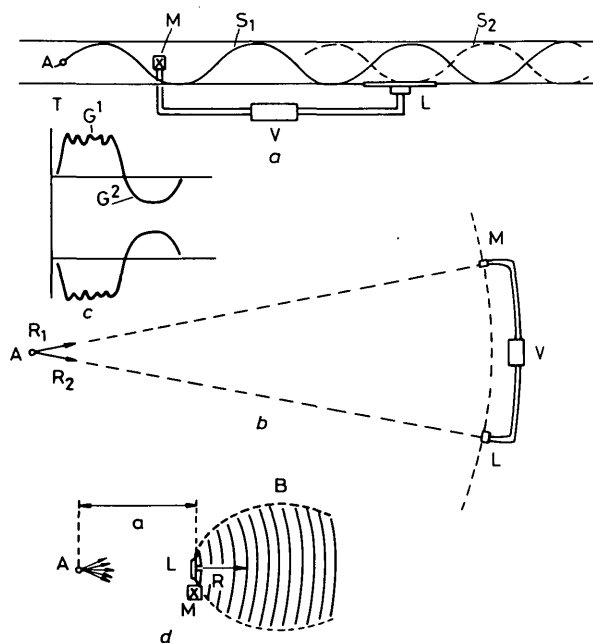


Fig. 1 Illustrations from Lueg's patent

In Fig. 1a Lueg shows the problem of cancelling sound in a duct. Here, acoustic noise propagates along the duct T through point A . The microphone M detects the sound and converts it to an electrical signal. The electrical signal is fed to amplifier V and then to loudspeaker L . Lueg shows a single frequency in the duct, the phase reversal of which is accomplished by considering the electronic system V as a transmission line with a given time delay. The length of the line is adjusted to give the necessary time delay that results in 180° phase shift, at a specified frequency, relative to the sound wave detected at the microphone, so that cancellation of a wave of that particular frequency results.

In Fig. 1b Lueg shows the problem of cancelling sound oscillations in a limited area near the loudspeaker. Here, the movement of the diaphragm of the loudspeaker is synchronised in antiphase with the noise arriving at

position L . The zone of attenuation of sound around the loudspeaker is a result of the sound energy absorption by the loudspeaker that occurs in this process. The third proposal made by Lueg is shown in Fig. 1d where he describes the case of noise attenuation in an open space of a point source of noise located at A . The microphone M detects the sound wave which, after amplification, is delivered to loudspeaker L to produce a cancelling wave in the direction of propagation R , resulting in a zone of silence B . In this case attenuation occurs by a process of destructive interference.

In Figs. 1a, b and d both the interference and absorption principles have been employed. In diagrams a and b the unwanted noise is detected and processed by the control system V such that by the time the noise reaches the loudspeaker location the movements of the diaphragm of the loudspeaker are in antiphase synchronisation with the noise, and so the loudspeaker acts as an absorber. The acoustic energy is absorbed by the loudspeaker producing cancellation beyond its location, in the direction of propagation. The fact that the loudspeaker acts as an acoustic energy absorber has been experimentally verified by the authors. Two loudspeakers, one as a primary source and one as a variable-phase secondary source, were driven by the same single-frequency source. When brought face to face and adjusted to be in antiphase a reduction in the electrical energy taken by the loudspeakers was detected. Although small, owing to the relative inefficiency of the loudspeakers, the effect was clearly demonstrated by varying the relative phase until no reduction was detectable. The system illustrated in Fig. 1d employs the principle of interference. Here the detected noise is delayed in phase by 180° and fed to the loudspeaker which superimposes it on the unwanted noise thereby producing destructive interference. The loudspeaker no longer absorbs the sound energy but, by superimposing the antiphase noise on the unwanted noise, redistributes the acoustic energy in the open space producing regions of cancellation and reinforcement.

In Fig. 1a Lueg correctly illustrates the basic physical phenomenon which provides the possibility of ANC for this case:

'An acoustic wave with a specific frequency has a relatively much lower speed than an electrical signal of the same frequency. This implies that while a sound wave is travelling from a point where it is detected to a point where it is to be attenuated, there is enough time available within the electronic circuit to process the signal and activate the control elements, to a greater or lesser degree, depending on the frequency, type of noise, and physical extent of the system'.

Note that this basic phenomenon is effective for a single frequency or very narrow band noise and, moreover, is effective if the relative distance of the detector from the primary source is less than that of the secondary source relative to the primary source on the same propagation side of the primary source. If the noise is of a broadband nature, the controller path of the system cannot be considered as a simple transmission line but is required to realise a continuous transfer function that phase shifts each detected component frequency, by 180° , relative to the noise.

In applying the above-mentioned physical phenomenon, Lueg shows (Fig. 1) that within the time interval required for the passage of an acoustic wave from the

detection point M to the control point L sound can be detected by the microphone, passed through the amplifier (controller) V and fed to the loudspeaker. In this process, if the acoustic wave does not change, it will be cancelled at position L provided the phase difference between the wave emitted by the loudspeaker and the unwanted acoustic wave is an odd multiple of 180° . If the controller path is of a given time delay cancellation of only a specific frequency will occur. Waves at other frequencies will be less well attenuated or even reinforced, depending on their frequency. This implies that for duct noise, as illustrated by Lueg, if the amplitude of the acoustic wave in travelling from the detection point to the control point remains unchanged, then for good cancellation it is necessary to take into consideration the delay time due to the passage of the acoustic wave from the detection point to the control point in relation to the phase-correction characteristics the controller is required to have so that cancellation occurs at the control point. In a more general case, however, the controller is to have the required phase-correction as well as amplitude-correction characteristics to obtain cancellation. Unfortunately, the electronic technology of the 1930s was not sufficiently advanced to meet the requirements of active noise attenuation systems, i.e. the phase- and amplitude-correction characteristics required of the controller, and practical results were not realised.

There does not appear to have been any further development by Lueg, and active noise attenuation rested for about 20 years until Olson introduced his 'electronic sound absorber' in 1953 [9]. In this and a later paper [10] Olson proposed localised sound reducers for occupants of vehicles and for machine operators, machinery noise control, noise-reducing headsets, duct noise reduction etc.

Olson's absorber is shown in Fig. 2. The microphone detects the unwanted noise and feeds it to the amplifier.

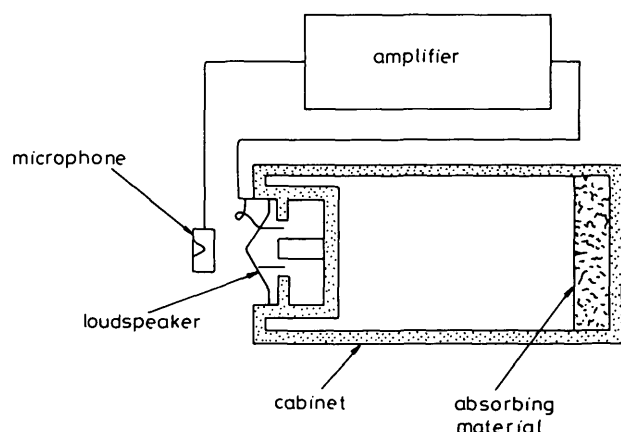


Fig. 2 Olson's electronic sound absorber

The amplifier, in turn, drives the loudspeaker such that the sound pressure at the microphone location is reduced. This effectively creates a 'zone of cancellation' in front of the absorber. As might be expected from the advance of technology in the 1950s, Olson's device shows considerable sophistication and ingenuity. In fact, it was possible to control the phase with good accuracy over a reasonably broad frequency range. However, the basic feedback structure of the system, along with other system errors, limits the utility of Olson's device.

The feedback path which includes the detector microphone, amplifier, loudspeaker and the spatial gap between the loudspeaker and the microphone (Fig. 2) is

required to have infinitely high gain to obtain good reduction in the unwanted noise level. Increasing the gain of the feedback path will cause the system to become unstable thereby limiting the performance of the system.

The performance of the Olson device as a 'sound pressure reducer' at the detector location is shown in Fig. 3.

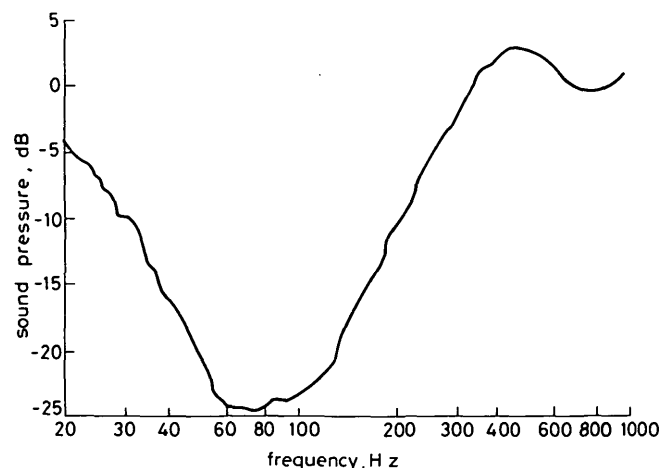


Fig. 3 Performance of Olson's electronic sound absorber at microphone location

The high-frequency performance of the system is limited by the feedback control structure of the system and the limited electronic capabilities, whereas the low-frequency performance limitation is due to the response of the loudspeaker in that range. Nevertheless, the performance of the device as measured at the microphone location is good. Unfortunately, as shown in Fig. 4, both the

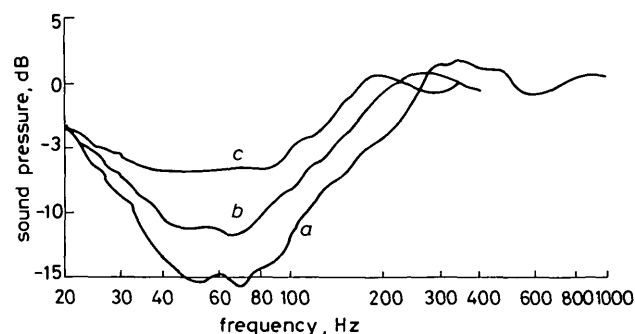


Fig. 4 Performance of Olson's device at various distances relative to microphone

a 10 cm (4 in)
b 25 cm (10 in)
c 61 cm (24 in)

amount and frequency range of attenuation decrease for increasing distances from the microphone, making the physical extent of attenuation of the device so limited that at distances of more than about 30 cm from the microphone the system is of no practical use. This is due to the interference pattern set up by the two sources redistributing the sound energy. An area of cancellation is produced around the loudspeaker but larger areas of reinforcement are produced at locations further from the source. There do not seem to be any further developments by Olson, however, his proposal has recently been investigated for the active control of sound in ducts [11, 12].

In 1955 Simshauser and Hawley [13] proposed the development of an 'active ear defender'; a device to reduce ambient noise at the ear by using a headset to generate a sound pressure equal in magnitude and

opposite in phase to the noise. The device is a two-channel system consisting of two microphones mounted one on each earphone of a conventional military headset with an amplifier and a phase-shifting network on each channel. Noise is detected by the microphone, passed through the amplifier and phase-shifting network and applied to its earphone. Simshauser and Hawley tested the device in a pure-tone field with proper amplitude and phase adjustments for minimum loudness. The results of their calculations, based on experimental measurements, indicated that such a device would provide an average reduction of 10 dB more than is provided by the earcap alone in the range 100 Hz to 1200 Hz. This is currently one of the more successful applications of ANC [14–17].

At about the same time as Olson and Simshauser, William Conover of General Electric discussed the active control of transformer noise [18]. His work was based on a relatively large, 15 MVA, transformer installation. Unlike Olson and Simshauser he did not confine his work to the laboratory but was rather successful in that he developed his work in the field! Conover's scheme was to place loudspeakers near the transformer's surface and cancel the pressure radiation in the near field. He argued that sound radiates from a transformer owing to vibrations of the core, caused principally by magnetostriction, and is transmitted through both the core mountings and the fluid cooling medium to the tank, which in turn couples these vibrations to the surrounding air. This produces a periodic spectrum with harmonics at even multiples of the supply frequency. In a 50 Hz transformer, vibration frequencies are multiples of 100 Hz, and the mode of the tank vibration will in general be different for each frequency. Furthermore, an important characteristic of transformer noise is that the first few harmonics are usually the most important contributors to the sound level. For example the 100, 200 or 300 Hz component is likely to outweigh the combined effect of the higher harmonics. Accordingly, a substantial reduction of one or more of the first three components is usually sufficient to achieve a reduction in the noise.

Conover used the 60 Hz supply to the transformer to generate the 120, 240 and 360 Hz harmonics of the fundamental frequency directly. This is a major advantage of this application as no acoustic detection is required and therefore acoustic feedback is eliminated. After adjusting each component in amplitude and phase, the signals were recombined, amplified and fed to a speaker placed at the centre of one of the flat faces of the transformer. Measurements at 30.5 m (100 ft) radial distance from the transformer showed a reduction of more than 6 dB within an angular zone of about 11.5° on either side of the line drawn radially perpendicular to the speaker face. Beyond this zone, as shown in Fig. 5, reinforcement took place. Owing to changes in the measured results over time, he suggested the design of a 'black box' controller to adjust the amplitude and phase of the cancelling signal in accordance with the actual measurements of the time-varying noise detected by a microphone placed at a distant position from the transformer-speaker combination.

Conover was the first to attempt the attenuation of transformer noise by active methods [18]. Since then this has become a classic problem investigated directly or indirectly by Kido [19], Hesselman [20], Ross [21], Jessel and Angevine [22, 23]. Much work has been devoted to the transformer noise problem because it is potentially so attractive. Transformer hum is annoying and pervasive; it is of economic importance and seems

readily solvable by active means because the frequencies generated are periodic in nature and precisely related to the supply. However, practical systems are still not available.

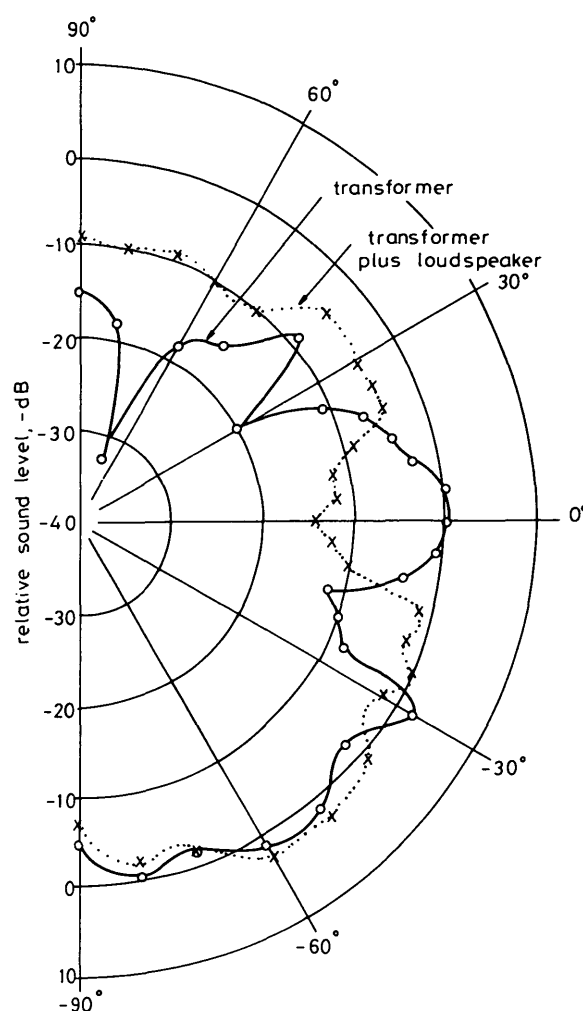


Fig. 5 Noise reduction of a 15 MVA transformer, by Conover

As depicted in Fig. 5, Conover's work resulted in a limited ray or beam of cancellation with the noise intensified in other areas. It is now clear [24] that the noise of a physical source may be reduced everywhere in the far field, to some degree, as long as the source is much smaller than the wavelength of the maximum frequency component λ_m of the radiated sound, and the cancelling source and radiating source are located less than $\lambda_m/3$ or $\lambda_m/4$ apart. When the size of the source is much less than the wavelength of the radiated sound, the vibration of the surface is generally in phase; in which case the combined acoustic pressures of the source and the out-of-phase cancelling signal tend to cancel each other near the source. In the case where the physical size of the source is large compared with the wavelength λ_m of the radiated sound, the surface no longer moves in a simple uniform phase relationship but rather in a more complicated pattern where the phase varies over different regions of such a physical source. As a consequence, the radiation pattern of the source is redistributed causing local reduction of the sound pressure without having much effect on the total radiated sound energy. Certain zones are thus produced in which the sound level is reduced whereas it is intensified in other locations.

Looking back to the early developments in ANC, Lueg's initial efforts were unsuccessful since no papers

were published following his patent and no applications are known in the 1930s and 1940s. In the 1950s, Olson, Simshauser and Conover showed considerable interest in active noise attenuation but their efforts were also unsuccessful. Therefore, interest again subsided in the late 1950s and early 1960s. In retrospect, this was due to a poor understanding of the problem from an acoustic and control systems point of view as well as to the lack of necessary technology to be able to deal with the main problems associated with the design of successful ANC systems, which are as follows:

- (i) broadband noise sources
- (ii) distributed noise sources
- (iii) time-varying noise sources
- (iv) acoustic response of the medium
- (v) characteristics of transducers and other electronic equipment used

The unwanted noise emitted by noise sources is generally spread over a band of frequencies, and so practical ANC systems are required to be able to cope with such broadband noise. This implies that the controller employed is to have frequency-dependent characteristics so that the phase and the amplitude of each frequency component can be adjusted such that, when superimposed on the source noise, the unwanted noise is attenuated over the broad frequency range of the noise spectrum. A constant-gain controller, such as an amplifier, can provide large amounts of cancellation only at a single frequency. It cannot be used as an efficient controller for broadband noise cancellation.

A feature of many noise sources is that they are not compact; noise is not emitted by one small part of the source but is distributed over the entire surface. A distributed noise source can be approximated by a number of compact sources (multiple sources) distributed around the surface, in which case the control problem is much more complicated than for a single compact source. Here, a single detector and secondary source may not be sufficient to attenuate the unwanted noise but rather a number of them, depending on geometry limitations, are required.

A further restriction on the success of ANC systems is that the characteristics of many practical sources vary with operating conditions and hence with time. Dealing with a time-varying source in an ANC system, the performance of a controller with fixed frequency-dependent characteristics is no longer satisfactory. Here the system is required to have the capability of changing the characteristics of the controller in accordance with a change in the noise pattern of the source; i.e. the system should be able to adapt to changes in the characteristics of the source.

An acoustic wave travelling through a propagation medium is changed in amplitude as well as in phase owing to the response characteristics of the medium, and so each component frequency of the noise emitted undergoes an amplitude and phase change from the point of emission to the point where it is detected. Moreover, acoustic feedback and reflected waves have a significant influence on the performance of an ANC system. These factors depend on the geometric arrangement of the ANC system and determine the stability of the system. An analysis of the acoustic response of a nondispersive (linear) propagation medium to free progressive sound and its effect on the performance of an ANC system has been carried out by the authors and is soon to be submitted for publication.

Consider a uniformly pulsating source placed in a linear propagation medium and emitting sound waves in all directions. If there are no boundaries or obstacles to reflect sound waves back towards the source, the sound waves in propagating through the medium are attenuated in accordance with the inverse square law. This results in a sound level in the medium that decreases for increasing distances from the source. The major factors affecting the performance of an ANC system in such a case are: (a) characteristics of the source, (b) the acoustic properties of the medium, (c) the geometrical arrangement of the system, and (d) the characteristics of the electronic equipment used. However, if the source emits waves into a finite enclosure, the performance of an ANC system, besides the above factors, is also affected by the acoustic response of the enclosure.

If a sound source having components extending over a band of frequencies radiates energy into a large, irregular enclosure, fluctuations in the sound pressure are observed as a microphone is moved around the enclosure. However, in this case the maxima and minima of sound pressure lie much closer together in position than for either a small or a rectangular enclosure. In the low-frequency range there will be a few room resonances whereas at high frequencies there will be many resonances in any given band of frequencies. As ANC is restricted to the reduction of low-frequency sound, only a limited number of resonances occur. Furthermore, a common assumption, to be detailed later, is that the primary source is compact (with respect to the frequencies of interest) and that the secondary source is physically close to the primary source, again with respect to the frequencies of interest.

Advances in technology which began prior to the mid-1960s, and which are continuing, make the active control of noise much more attractive and feasible than it has ever been before. The areas of technology where the most important applicable progress has occurred are: advanced control systems technology, which includes the development of adaptive algorithms; improved understanding of the physics of acoustic systems; and the development of solid-state electronics, which makes sophisticated control realisable and inexpensive.

The accelerating interest in active control of noise began in the late 1960s with the publications of Jessel and his co-workers in France, and Kido and his associates in Japan. Jessel's work has been mainly concerned with duct noise [25, 26]. He and his co-workers have also made significant contributions to the theory of active attenuators [22, 27, 28] largely based on the development of Huygens' Principle, Kido's work, on the other hand, has been concerned with transformer noise [19].

It was realised rather early by both Jessel and Kido that the primary advantage of ANC systems is in their ability to attenuate low-frequency noise. This is an area of considerable interest because of the pervasiveness of low-frequency sources and the high cost, large bulk and relative inefficiency of current passive hardware in low-frequency applications, especially with regard to transmission loss and silencing [24, 29]. Besides this, an advantage in the control of the one-dimensional propagation of duct noise lies in the fact that active duct noise silencers produce no back pressure.

Jessel, and others, also discovered some of the problems associated with reducing duct noise. Longitudinal duct modes leading to acoustic feedback, due to reflected components, tend to confuse the controllers as to the exact level of the noise itself, since the detector micro-

phone cannot distinguish between the noise and the reflected components. This leads to system instability and/or no noise reduction in some bands of frequency. To solve the longitudinal-mode problem, so that the detector microphone detects the unwanted noise only, loudspeaker arrays can be used. Two or three loudspeakers used to form an acoustic dipole or an acoustic tripole can be phased to produce acoustic waves travelling in one direction. Two or three microphones in arrays, i.e. microphone arrays, may be used to obtain directional sensing characteristics which also tend to reduce the longitudinal-mode problem.

There has been considerable effort devoted to duct noise since Lueg's invention and primarily since Jessel's first work [11, 12, 25, 26, 30–72]. These efforts have resulted in the development of various systems, depicted in Fig. 6, for duct noise. The acoustic monopole system is

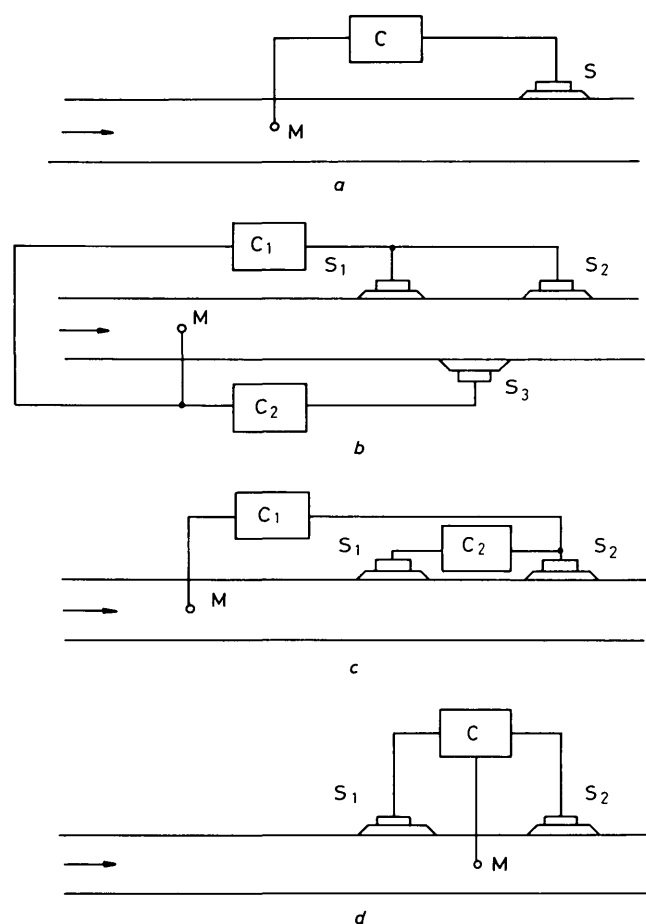


Fig. 6 Types of active noise attenuators for duct noise

- a Monopole (Lueg)
- b Tripole (Jessel)
- c Dipole (Swinbanks)
- d Dual monopole (Chelsea system)

that originally considered by Lueg. In this system, as the secondary (cancelling) source radiates upstream and downstream, a standing wave is produced upstream which interferes with the detection of the unwanted noise. This makes the position of the microphone extremely sensitive to noise reduction. The acoustic tripole and the acoustic dipole were developed by Jessel and his co-workers [25] and Swinbanks [30], respectively. They attempt to provide a cancelling signal in the duct that propagates only in the downstream direction. The Chelsea System, developed by Leventhal [31], is formed by two secondary sources with the detector midway between them. The controller is set to null the resultant

of the secondary sources' waves at the detector location thus isolating the detector from secondary-source radiation. The performance of these systems [31, 37] shows that they provide noise cancellation of up to 20–25 dB over a narrow band of less than an octave.

Multiple loudspeaker systems, i.e. dipole and tripole, have essentially been attempts to overcome the frequency-dependent problems of controller design. However, they have been unable to cope with reflected waves and have geometry-related limitations. The control problem is also much more complex in such systems. The third limitation of these systems is the 'tuning effect' due to the physical spacings of the microphone and loudspeakers relative to each other. By altering these spacings the system is tuned to a different frequency with no significant improvement in the bandwidth of attenuation. For these reasons, the current trend is to use the capabilities of modern electronic systems to implement complex controllers and hence give preference to controller complexity over geometrical complexity [26, 37, 42, 44, 46, 49, 53, 54, 58, 66, 67]. With proper control the monopole is capable of broadband high-attenuation performance, and, moreover, the control problem is simplified with a single source. However, the task of high attenuation over a broad band primarily lies in the design of the controller employed. What is required, therefore, is a proper ANC system design procedure based on sound analytical conditions for general phase cancellation [73].

Like the problem of ANC in ducts, a considerable amount of effort has been devoted to the control of noise in three dimensions, the greatest attraction being towards transformer noise since Conover's early work [19–21, 23, 74] and more recently towards motor vehicle noise [75–81]. However, some other specific applications have also been reported [82–90] as well as theoretical contributions towards understanding the general problem [27, 28, 69, 73, 89, 91–96]. But there has been little consideration of general three-dimensional ANC systems [57, 73, 83, 87].

The application of modern control concepts and acoustic theory to the control of transformer noise has revealed that a significant amount of reduction, 10 to 30 dB, in the fundamental single-frequency component can be obtained everywhere in the far field provided the physical source is smaller than the wavelength of the radiated sound, and the cancelling source is located near the radiating source. When the physical source is larger than the wavelength of the radiated sound and/or for harmonics of the fundamental frequency component, only certain limited zones of cancellation can be produced. The size of these zones can be increased by the addition of more detectors and loudspeakers. This redistributes the sound energy over the space dividing the region into zones of cancellation and sound intensification. The situation can be improved by matching an array of detectors and cancellation sources with the subdivided space so that one element of the array coincides with each subdivision. This method has clear limitations in complexity and cost as well as design. The addition of each element to the array improves the attenuation in the sense that the zones of cancellation are increased such that the sound energy distribution pattern is changed thereby creating new zones of cancellation and sound intensification. This means that the problem of eliminating the sound intensification zones will still remain unsolved.

The problem of time-varying noise sources is another dimension of ANC systems the designer has to consider.

In fact, it is an essential requirement for many sources and is the crucial difference between laboratory success and practical systems. Here a fixed control system is of limited use in attenuating the noise because of the continual adjustment required in the system as the noise pattern of the source varies from time to time. One approach to this problem is to use the concepts of adaptive control systems. Implementing an adaptive control algorithm within the ANC system will allow the system to vary the controller characteristics in accordance with changes in the noise pattern of the source. Recently, Chaplin and his co-workers [39, 78, 79, 81, 98–100] have reported some success with an adaptive method based on trial and error waveform generation. This method, however, relies on the noise source being periodic. There is no doubt that the implementation of a more general adaptive controller, based on a stochastic parameter estimation algorithm, is a potentially attractive solution to reducing the noise emitted by a time-varying broadband source. Recently, some theoretical and practical investigations have been reported in this area [70, 101–105].

It is noted that in most of the papers on ANC very little discussion is devoted to the electronic systems used in achieving the results obtained. Also, frequency-domain and control system concepts are only rarely used in the analysis and design of ANC systems [73, 87, 106, 107]. Similarly, little is found about geometrical considerations in the literature. Details of the best arrangements are not given. Analytical approaches, on the other hand, which have treated the problem mathematically have always been extremely general because of the highly complicated nature of the problem. Thus, details suitable for design purposes have not been developed. Nevertheless, the physical geometry has been shown to be extremely important in achieving high attenuation [42]. A high degree of attenuation can be expected only with geometries which favour proper mixing of the cancelling sound field with the unwanted noise field.

The field of ANC is attracting considerable worldwide attention. There are programmes continuing in Britain, France, Germany, Japan, Poland, Russia and the USA. There is hope that the rapid development of this field will result in successful applications in ANC in the years to come. This is because of the advantages and capabilities of ANC systems as compared with passive methods; namely (i) considerably improved low-frequency performance, (ii) reduction in hardware cost for many applications, (iii) great reduction of size and weight, and (iv) zero, or very low, back pressure.

The old concept of active noise attenuation is becoming a reality outside the laboratory because new technological developments are providing the opportunity for the advantages to be realised. Owing to the current developments in technology, it seems reasonable to expect, in the near future, commercially available hybrid active-passive noise attenuators [24], for duct noise, in which the low-frequency attenuation, below 500 Hz, is provided by an active system and the high-frequency attenuation is provided by passive hardware.

The main part of an ANC system which needs to be given greater consideration is a design procedure for the controller. Because of the broadband nature of many acoustic sources, the controller is required to have frequency-dependent characteristics. The dependence of these characteristics on a number of frequency-dependent factors within the system, such as the characteristics of the source, transducers and other electronic equipment, and the propagation medium, makes it possible to

measure and realise the required characteristics as a controller. In the past, because of the complexity involved in implementing such a controller, geometrical complexity, by using a number of secondary sources, was preferred. However, with the capabilities of current electronic technology, especially digital electronics, it is possible to implement such a controller either as a fixed digital controller or as an adaptive digital controller, whichever is required by the application, on a special-purpose micro-processor.

3 General conditions for phase cancellation

Active cancellation of unwanted noise is based on the intentional superposition of acoustic waves such that destructive interference results. This uses an artificially generated source to emit sound waves that interfere with the unwanted noise so that the destructive interference of the component waves can be used to result in the attenuation of the unwanted noise. There are two factors that influence the cancellation of the unwanted noise, namely the artificially generated (secondary) source as related to the unwanted (primary) source, and the properties of the medium into which both sources emit waves.

To achieve cancellation at a given point in the propagation medium it is required that the two sources be coherent. This implies that the secondary source must emit a wave of the same frequency as the primary wave and with a constant phase relative to it. The second requirement in this process is that the two waves be in antiphase at the cancellation point so that they interfere with each other destructively. In the absence of the first requirement the two waves are uncorrelated and stable cancellation is not possible. If the first requirement is fulfilled but not the second, the interference between the waves may be constructive implying a reinforcement of the primary wave and an overall increase in sound intensity.

In propagating through the medium the waves are affected by the properties of the medium. The requirement of coherency implies that the medium should be linear. After propagating in the medium from the sources to a specified point, the effect of the medium on the amplitudes as well as the phases of the component waves will determine whether the interference is destructive or constructive. Here the separation between the sources plays a major role. Moreover, geometry-related conditions, involving the separation between the sources and the location of a point in the medium at which cancellation is required, are essential. These conditions are expressed, for a point source emitting free progressive waves into a linear propagation medium, in this Section in a format which provides a systematic method for the design of ANC systems.

3.1 Field cancellation factor

Let a (primary) source S_p be placed in a nondispersive (linear) propagation medium and emit a (sound) wave $p(t)$ as a function of time t . It produces a field $p(r, t)$ in the medium, where r is the distance measured from S_p . At any fixed point O at a distance r_p from S_p the wave $p(t)$ will give rise to a field $p(r_p, t)$, or $p_o(t)$, which is a function of time only. As depicted in Fig. 7, let a second source S_s be placed at a distance d from S_p in the medium and emit a wave $s(t)$. In propagating through a distance r_s from S_s , the wave $s(t)$ will give rise to $s_o(t)$ at point O . The combination of the component waves from S_p and S_s at point O will result in a signal $o(t)$. With ω representing the

radian frequency, let

$G_{pp}(\omega)$ = autopower spectral density of $p(t)$
 $G_{ss}(\omega)$ = autopower spectral density of $s(t)$
 $G_{ppo}(\omega)$ = autopower spectral density of $p_o(t)$
 $G_{sso}(\omega)$ = autopower spectral density of $s_o(t)$
 $G_{cco}(\omega)$ = autopower spectral density of $o(t)$.

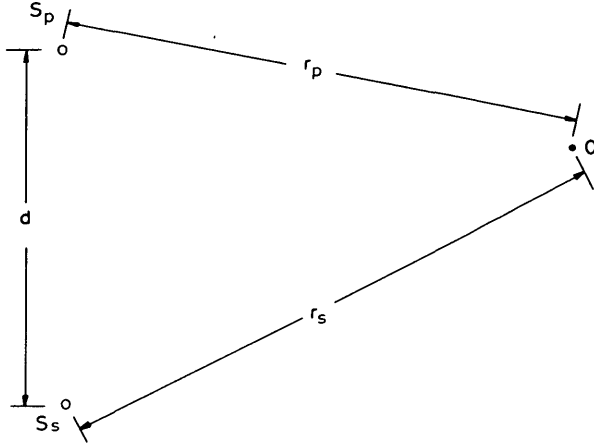


Fig. 7 Primary and secondary sources emit acoustic waves to produce cancellation at point O

Cancellation at the observation point O occurs when the power spectral density of the combined signal is less than the primary signal acting alone, giving

$$G_{cco}(\omega) < G_{ppo}(\omega) \quad (1)$$

For the purpose of a quantitative description of the phenomenon of cancellation we shall define the field cancellation factor K as the ratio of the cancelled spectrum $G_{ppo}(\omega) - G_{cco}(\omega)$ to the primary spectrum $G_{ppo}(\omega)$ which existed at the given point O prior to the superposition of the secondary field $s_o(t)$; i.e.

$$K \triangleq \frac{G_{ppo}(\omega) - G_{cco}(\omega)}{G_{ppo}(\omega)} \quad (2)$$

or

$$K = 1 - \frac{G_{cco}(\omega)}{G_{ppo}(\omega)} \quad (3)$$

Using the cancellation requirement, eqn. 1, in eqn. 3 it follows that for cancellation to occur the field cancellation factor must lie between zero and unity:

$$0 < K \leq 1 \quad (4)$$

A zero K corresponds to no cancellation and a unity K corresponds to complete cancellation.

The basic system of Fig. 7 can be drawn as in Fig. 8 where $X_p(j\omega)$ and $X_s(j\omega)$ represent the transfer characteristics of the medium from S_p and S_s , respectively, through the distances r_p and r_s to the observation point O , and are

$$X_p(j\omega) = H_p \exp(-j\omega t_p) \quad (5a)$$

$$X_s(j\omega) = H_s \exp(-j\omega t_s) \quad (5b)$$

where t_p and t_s are the times by which the primary and secondary waves are delayed in propagating through the distances r_p and r_s from S_p and S_s , respectively, to the observation point O . These delay times are expressed as

$$t_p = \frac{r_p}{c} \quad t_s = \frac{r_s}{c} \quad (6)$$

where c is the velocity of sound in the medium. The attenuating factors H_p and H_s , on the other hand, are inversely proportional, respectively, to the square of the distances r_p and r_s .

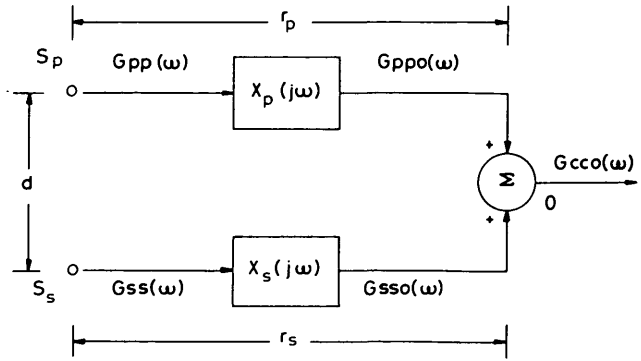


Fig. 8 Transfer function description of acoustic paths

If the primary signals $p(t)$ and $p_o(t)$ are represented by $P(j\omega)$ and $P_o(j\omega)$, respectively, and the secondary signals $s(t)$ and $s_o(t)$ are represented by $S(j\omega)$ and $S_o(j\omega)$, respectively, in the frequency domain then

$$P_o(j\omega) = X_p(j\omega)P(j\omega) \quad (7a)$$

$$S_o(j\omega) = X_s(j\omega)S(j\omega) \quad (7b)$$

and the autopower spectral densities can be written as [108, 109]

$$G_{pp}(\omega) = |P(j\omega)|^2 \quad (8a)$$

$$G_{ss}(\omega) = |S(j\omega)|^2 \quad (8b)$$

Using eqns. 5, 7 and 8 it follows that the spectral densities of $p_o(t)$, $s_o(t)$ and $o(t)$ are

$$G_{ppo}(\omega) = |P_o(j\omega)|^2 = H_p^2 G_{pp}(\omega) \quad (9a)$$

$$G_{sso}(\omega) = |S_o(j\omega)|^2 = H_s^2 G_{ss}(\omega) \quad (9b)$$

and

$$G_{cco}(\omega) = |P_o(j\omega) + S_o(j\omega)|^2 \\ = |X_p(j\omega)P(j\omega) + X_s(j\omega)S(j\omega)|^2$$

which simplifies to

$$G_{cco}(\omega) = G_{ppo}(\omega) + G_{sso}(\omega) \\ + 2\sqrt{G_{ppo}(\omega)G_{sso}(\omega)} \cos \left[\omega \frac{\Delta r}{c} - \theta(\omega) \right] \quad (10)$$

where $\Delta r/c = (r_p - r_s)/c = t_p - t_s$, and $\theta(\omega)$ is the phase by which $p(t)$ leads $s(t)$.

Substitution of $G_{cco}(\omega)$ from eqn. 10 into eqn. 3, after simplification, yields

$$K = - \frac{G_{sso}(\omega)}{G_{ppo}(\omega)} - 2 \sqrt{\frac{G_{sso}(\omega)}{G_{ppo}(\omega)}} \cos \left[\omega \frac{\Delta r}{c} - \theta(\omega) \right] \quad (11)$$

If the ratio of the spectral densities of the secondary wave to the primary wave, or, simply, the ratio of the secondary wave power to the primary wave power, is denoted by α ,

$$\alpha = \frac{G_{sso}(\omega)}{G_{ppo}(\omega)} \quad (12a)$$

and the cross-spectral density factor defined by the cosinusoidal factor in eqns. 10 and 11 is denoted by β ,

$$\beta = \cos \left[\omega \frac{\Delta r}{c} - \theta(\omega) \right] \quad (12b)$$

then eqn. 11 can be written as

$$K = -2\beta\sqrt{\alpha} - \alpha \quad (13)$$

This gives an analytical relationship between the cross-power spectral density factor, interpreted as the relative phase, the autopower spectral density ratio, interpreted as the relative amplitudes, and the degree of cancellation given by the cancellation factor K .

3.2 Conditions for cancellation

Based on the results obtained in the preceding Section, the general conditions for field cancellation can now be obtained. Substituting the value of K from eqn. 13 into eqn. 4 and simplifying yields

$$\frac{\sqrt{\alpha}}{2} < -\beta \leq \frac{1+\alpha}{2\sqrt{\alpha}} \quad (14)$$

Since α is the ratio of powers of the waves it is therefore a positive real number. This implies that both the left- and right-hand sides of eqn. 14 are positive. Thus, condition 14 is only satisfied if the cross-spectral density factor assumes negative values; i.e.

$$\beta < 0 \quad (15)$$

This is the phase condition for cancellation and is shown in Fig. 9.

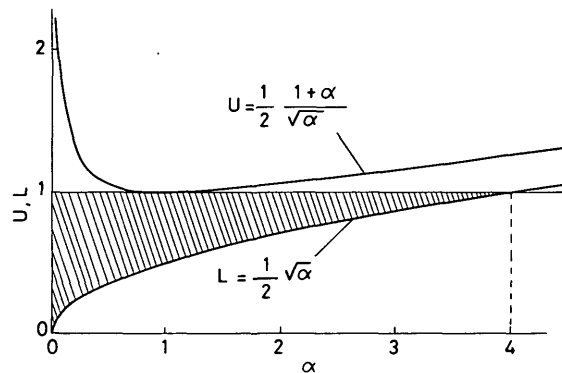


Fig. 9 Shaded region represents $-\beta$ values for which cancellation occurs for a certain α

The amplitude condition for cancellation can also be obtained from eqn. 14. As shown in Fig. 9, $-\beta$ lies between the functions U and L . However, from the definition of β , the cross-spectral density factor is bounded by one; i.e.

$$|\beta| \leq 1 \quad (16)$$

and, therefore, eqn. 14 becomes

$$\frac{\sqrt{\alpha}}{2} < -\beta \leq 1 \quad (17)$$

Thus the required amplitude condition for cancellation, shown by the shaded region in Fig. 9, is obtained as

$$0 < \frac{\sqrt{\alpha}}{2} < -\beta \quad (18)$$

Squaring this inequality, the range of the permissible power ratio is obtained to be

$$0 < \alpha < 4\beta^2 \quad (19)$$

Substitution of the value of α from eqn. 12a into eqn. 19 yields

$$0 < G_{ss}(\omega) < 4\beta^2 G_{pp}(\omega) \quad (20)$$

Note that if $-\beta$ is at its upper limit (unity) then eqn. 20 implies that cancellation at the observation point can occur if the spectral density of the secondary wave is less than four times the spectral density of the primary wave or, equivalently, if the power associated with the secondary wave does not exceed four times the power associated with the primary wave.

If β is treated as a parameter in eqn. 13, the field cancellation factor K as a function of power ratio α can be represented as a family of curves, each curve corresponding to a particular value of β , as shown in Fig. 10.

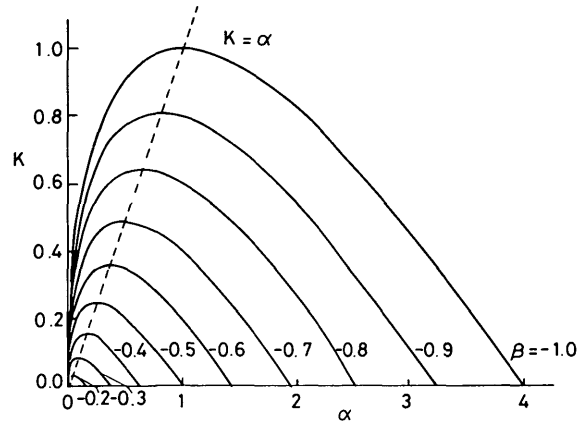


Fig. 10 Field cancellation factor as related to the power ratio and cross-spectral density factor of primary and secondary waves

Maximising the field cancellation factor in eqn. 13 with respect to the power ratio α it follows that maximum cancellation is obtained if

$$\alpha = \beta^2 \quad (21)$$

The corresponding maximum value of the field cancellation factor K_{max} is therefore

$$K_{max} = \beta^2 \quad (22)$$

From eqns. 21 and 22 it is clear that the maxima of the family of curves corresponding to eqn. 13, Fig. 10, lie on the straight line

$$K_{max} = \alpha \quad \text{for } \alpha \leq 1 \quad (23)$$

Solving eqn. 13 for $-\beta$ and substituting into eqn. 17 yields, after simplification,

$$0 < K \leq 2\sqrt{\alpha} - \alpha \quad (24)$$

The solution of the right-hand side of this inequality for α in terms of K reveals that cancellation larger than any K is possible for waves for which the power ratio α is within the limits

$$[1 - \sqrt{1 - K}]^2 \leq \alpha \leq [1 + \sqrt{1 - K}]^2 \quad (25)$$

and the magnitude of β is higher than $\min|\beta|$, where $\min|\beta|$, from a maximisation of eqn. 13 for β as a function of α , is

$$\min|\beta| = \sqrt{K} \quad (26)$$

The conditions of cancellation as related to the relative time delay between the primary and secondary waves reaching the observation point O and as related to the regions of cancellation are now to be considered.

The difference in distance Δr from the two sources to the point O as noted in eqn. 10 is

$$\Delta r = (t_p - t_s)c$$

or

$$\Delta r = \tau c \quad (27)$$

where τ is the relative time delay between primary and secondary waves, of the same frequency, reaching point O . Substituting the value of β from eqn. 12b into eqn. 17 and using eqn. 27 yields

$$\frac{\sqrt{\alpha}}{2} < -\cos [\omega\tau - \theta(\omega)] \leq 1 \quad (28)$$

from which it is clear that $\omega\tau - \theta(\omega)$ represents an angle either in the second or third quadrant, including the π axis. Solution of eqn. 28 for $\omega\tau - \theta(\omega)$ yields that for the secondary field to cancel the primary field the phase difference $\omega\tau - \theta(\omega)$ between the same-frequency components of the waves is required to be

$$(2n+1)\pi - \cos^{-1} \frac{\sqrt{\alpha}}{2} < \omega\tau - \theta(\omega) < (2n+1)\pi + \cos^{-1} \frac{\sqrt{\alpha}}{2} \quad (29)$$

where $n = 0, 1, 2, \dots$. This is shown in Fig. 11.

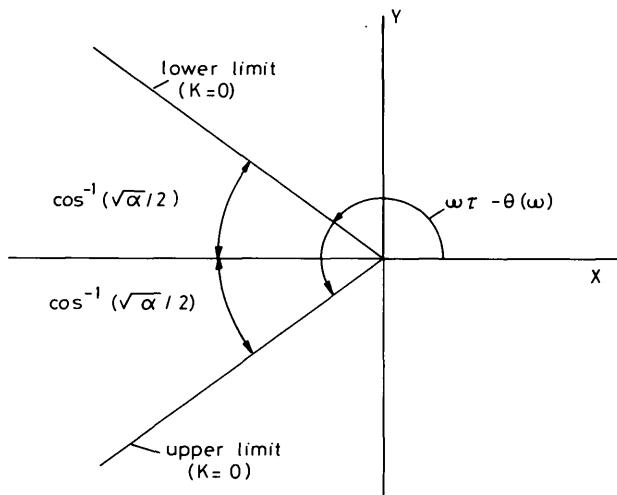


Fig. 11 Range of difference in phase between primary and secondary waves for cancellation

Using the relationship between the velocity of sound, the signal wavelength and the signal frequency, eqn. 27 can be written as

$$\omega\tau = \frac{2\pi}{\lambda} \Delta r \quad (30)$$

Substituting the value of $\omega\tau$ from eqn. 30 into eqn. 29 and simplifying, the following is obtained:

$$\left[(2n+1) + \frac{\theta(\omega)}{\pi} - \frac{1}{\pi} \cos^{-1} \frac{\sqrt{\alpha}}{2} \right] \frac{\lambda}{2} < \Delta r < \left[(2n+1) + \frac{\theta(\omega)}{\pi} + \frac{1}{\pi} \cos^{-1} \frac{\sqrt{\alpha}}{2} \right] \frac{\lambda}{2} \quad (31)$$

This implies that the range of cancellation for a specific frequency of the component waves is related in direct proportion to the wavelength, or, in other words, in inverse proportion to the frequency.

Substituting the value of β from eqn. 12b into eqn. 13 and using relations 27 and 30 yields

$$K = -2\sqrt{\alpha} \cos \left[\frac{\Delta r}{\lambda/2} \pi - \theta(\omega) \right] - \alpha \quad (32)$$

Maximum cancellation is obtained at points in the medium for which

$$\cos \left[\frac{\Delta r}{\lambda/2} \pi - \theta(\omega) \right] = -1$$

or

$$\Delta r = \left[(2n+1) + \frac{\theta(\omega)}{\pi} \right] \frac{\lambda}{2} \quad n = 0, 1, 2, \dots \quad (33)$$

Note that if $\theta(\omega) = 0$, eqn. 33 implies that for a particular frequency component of the waves maximum cancellation occurs at such points in the medium for which the difference in their distances from the two sources is exactly an odd multiple or half the wavelength, whereas if $\theta(\omega) = \pi$, maximum cancellation occurs at such points in the medium for which their distance differences from the sources are even multiples of half the wavelength. In the vicinity of these points there will also be cancellation which will decrease with the cosine of the ratio of distance difference divided by half the wavelength, symmetrically for distance differences smaller and larger than that in eqn. 33.

At points in the medium for which eqn. 33 holds and $K = K_{max}$, eqn. 32 gives the power ratio α as

$$\alpha = (1 \pm \sqrt{1 - K_{max}})^2 \quad (34)$$

confirming the fact that, for cancellation, α is to be limited between 0 and 4.

For any frequency component of the primary and secondary waves $p_o(t)$ and $s_o(t)$ with amplitudes P_p and S_o , respectively, the power ratio is

$$\alpha = \frac{S_o^2}{P_p^2} \quad (35)$$

Substitution of this relation into eqn. 19 yields that for the same-frequency components of the waves to cancel each other their amplitude ratio is required to be

$$0 < \left(\frac{S_o}{P_p} \right) < 2 \quad (36)$$

whereas from eqns. 34 and 35 it is revealed that the maximum possible cancellation is obtained ($K_{max} = 1$) for a unity amplitude ratio; i.e.

$$\left(\frac{S_o}{P_p} \right) = 1 \quad \text{for maximum cancellation} \quad (37)$$

From eqn. 35 it is clear that for given primary and secondary waves with known amplitude ratios, the power-related general conditions of cancellation can also be obtained working directly with the amplitude ratio.

The above analysis gives a quantitative description of the degree of cancellation in terms of the amplitude and phase parameters; namely the power ratio α or, equivalently, the amplitude ratio and the relative phase, or crosscorrelation factor β , of the primary and secondary waves and provides the design basis of ANC systems.

3.3 Experimental verification

The various theoretical results obtained in the preceding Sections were verified using two Isobarik loudspeakers, one as the primary source and one as the secondary source. The experimental arrangement is shown in Fig. 12. The equipment used in the experiment, alongside the two loudspeakers, are a dual-channel Solartron SP1200 signal processor, two 10k low-power Helipot poten-

tiometers, one NAIM power amplifier and a condenser-type Cirrus microphone.

The signal processor [110] provides a range of time-domain and frequency-domain analysis functions and

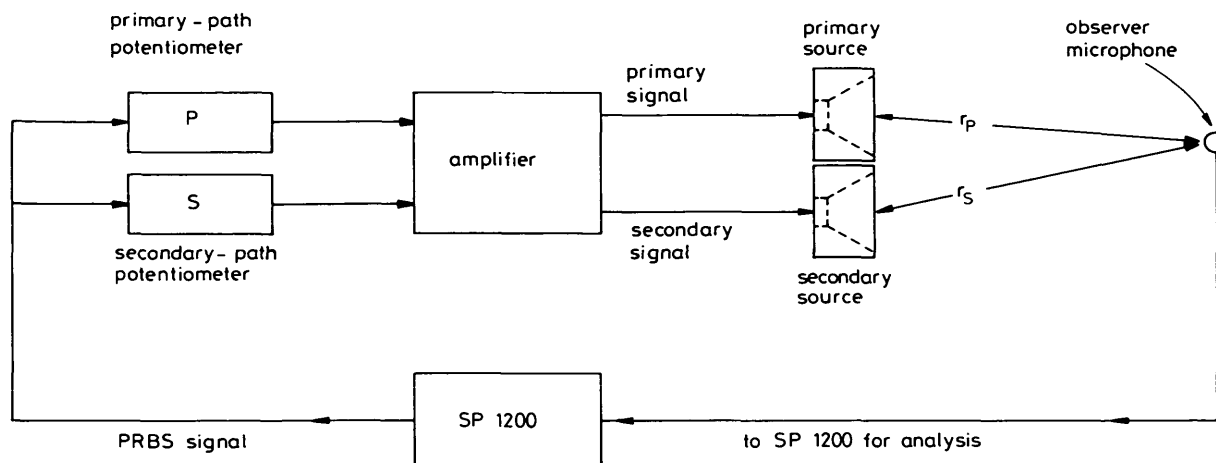


Fig. 12 Experimental set up for verification of general conditions of active noise cancellation

facilities for single- and dual-channel measurements. The analytical procedure of the device is based on the fast Fourier transform (FFT) through which the normal frequency-domain analysis functions, such as autopower, synchronous spectrum, crosspower, transfer function, coherence and power cepstrum with a maximum frequency of 30 kHz, are provided. The time-domain analysis functions provided by the instrument, via inverse FFT, are the time waveform, impulse response, autocorrelation and crosscorrelation functions. The instrument generates three types of test signals, namely PRBS (pseudo-random binary sequence) signal, periodic multi-sine signal and impulse signal. Among the input/output facilities of the instrument are two analogue input channels, digital plotter, analogue plotter and video signal output. The instrument's interface unit is the IEEE Standard 488 (1978) with full remote control.

The amplifier used is a quality stereo audio power amplifier. The amplitude characteristics of one of the channels has a constant offset of 0.2 dB relative to the second channel, but the phase characteristics of the two channels of the amplifier are within $\pm 0.1^\circ$ of each other. The amplitude characteristic of the amplifier is approximately flat, at an average value of 29.2 dB, above 20 Hz. The phase characteristic is $+6^\circ$ at 40 Hz, falls off to 0° at 400 Hz and stays at this level until 600 Hz above which it decays almost linearly, reaching -6° at 5 kHz. The amplifier does not have a variable gain on either of its channels; this was achieved by using two 10 k low-power Helipot potentiometers in the amplifier path.

A comparison of the transfer characteristics of the loudspeakers is shown in Fig. 13. The two amplitude characteristics are identical to within $+0.3$ and -0.9 dB in the range 0–900 Hz. Above 900 Hz and below 4.5 kHz they agree to within ± 3.7 dB. Above 4.5 kHz the characteristics diverge. The phase characteristics follow the same general trend as the amplitude characteristics. Both speakers are secured inside wooden frames, thereby reducing their back-pressure effects, and have a maximum cone diameter of 18 cm. If placed side by side the minimum separation between the centres of their cones is 34 cm.

Fig. 14 shows the transfer characteristics of the secondary source plus the condenser microphone. In measuring these characteristics the arrangement shown in

Fig. 15 has been used, indicating that they also include the acoustic path between the microphone and the loudspeaker cone, giving rise to the linear variation of phase with respect to frequency, and also the characteristics of

the amplifier used to amplify the microphone's output. The amplifier does not have a significant effect on the overall transfer characteristics of Fig. 14, whereas the effect of the acoustic path on the phase is clear from Fig. 14. Within the range 60–900 Hz the loudspeaker-microphone combination has an almost flat transfer characteristic. In the range 900–2800 Hz the amplitude characteristic varies between -3.5 and $+3.5$ dB. This variation reflects on the phase characteristic as a variable phase with an average equal to the linear phase characteristic resulting from the time delay of the acoustic path. The loudspeaker response diminishes above 4.2 kHz.

The experimental arrangement shown in Fig. 12 follows the basic theoretical structure of active noise control shown in Fig. 8 with the primary wave inverted ($\theta(\omega) = \pi$) and fed to the secondary loudspeaker to produce a secondary wave identical to the primary wave but delayed in phase by 180° relative to it.

Measurements of the primary signal's spectral density $G_{ppo}(\omega)$ and the spectral density of the combination of primary and secondary waves $G_{cco}(\omega)$ at an observation point, to form the ratio $G_{cco}(\omega)/G_{ppo}(\omega)$ in eqn. 3, can clearly show the sinusoidal variation of cancellation as a function of frequency. The results of such an experiment, using the SP 1200's PRBS signal over a frequency range of 0–5 kHz, are given in Fig. 16. At very low frequencies the sinusoidal shape of cancellation has been distorted owing to the difference in loudspeakers' characteristics in that region. At high frequencies cancellation and reinforcement diminish because of the attenuation and mismatch in loudspeakers' characteristics.

Using eqns. 12b and 13, the limit of cancellation ($K = 0$) is obtained to be at such frequencies for which

$$\omega = \frac{c}{\Delta r} \left[(2n + 1)\pi - \theta(\omega) \pm \cos^{-1} \frac{\sqrt{\alpha}}{2} \right] \quad n = 0, 1, 2, \dots \quad (38)$$

And if, specifically, $\theta(\omega) = \pi$, the frequencies $\omega_L = 2\pi f_L$ corresponding to the limit of cancellation are

$$\omega_L = \frac{2\pi n c}{\Delta r} \pm \frac{c}{\Delta r} \cos^{-1} \frac{\sqrt{\alpha}}{2} \quad n = 0, 1, 2, \dots \quad (39a)$$

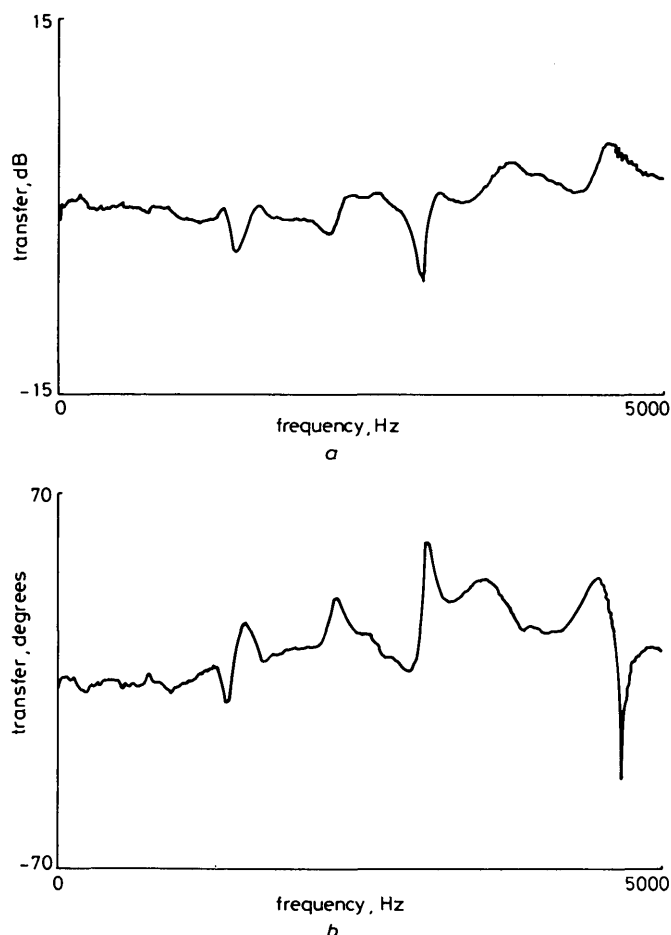


Fig. 13 Transfer characteristics of primary source relative to secondary source
a Amplitude
b Phase

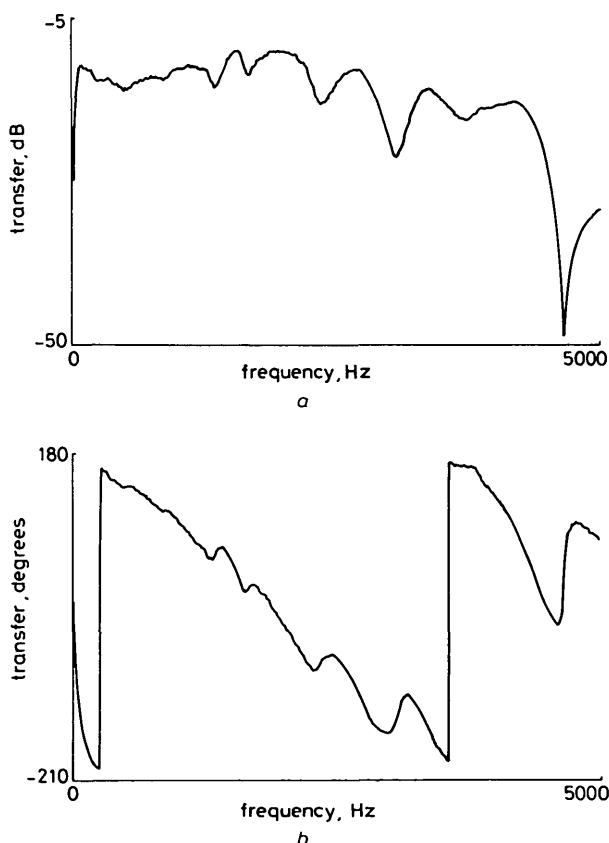


Fig. 14 Transfer characteristics of secondary source plus microphone
a Amplitude
b Phase

or

$$f_L = f_{mn} \pm \frac{c}{2\pi\Delta r} \cos^{-1} \frac{\sqrt{\alpha}}{2} \quad n = 0, 1, 2, \dots \quad (39b)$$

where

$$f_{mn} = \frac{nc}{\Delta r} \quad (40)$$

represents the frequencies (in Hz) at which maximum cancellation occurs.

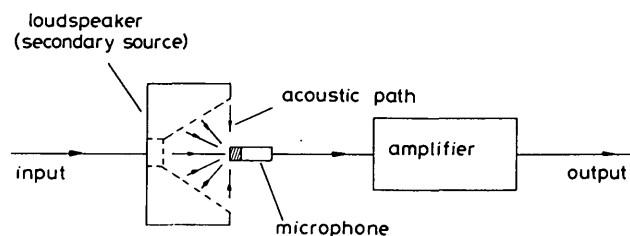


Fig. 15 Arrangement to measure transfer characteristics of loudspeaker plus microphone

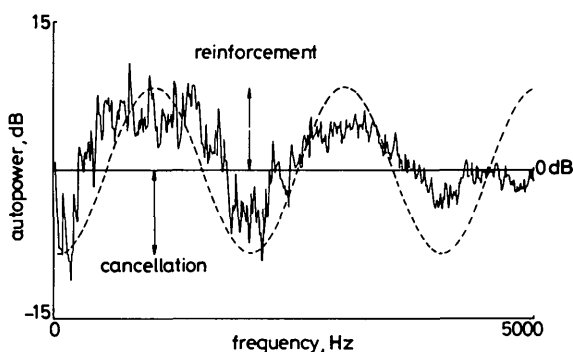


Fig. 16 Field cancellation as a function of frequency

To verify eqns. 39*a* and *b* the same experimental arrangement as in Fig. 12 was used. The observer microphone was placed opposite to the primary source with $r_p = 2m$ and the separation between the sources was varied so that $\Delta r = 0.385m$ was obtained at $d = 1.3m$. The sources were driven by a PRBS signal and adjusted so that a power ratio α of -6.0 dB was obtained over a range of 0–5 kHz at the observation point. Substituting these parameters and $c = 340$ m/s in eqns. 39 yields the frequencies of the limit of cancellation as

$$f_L = 883n \pm 185 \text{ Hz} \quad n = 0, 1, 2, \dots \quad (41)$$

and the frequencies at which maximum cancellation occurs as

$$f_{mn} = 883n \text{ Hz} \quad n = 0, 1, 2, \dots \quad (42)$$

Measuring the cancellation as a function of frequency, over a range of 0–5 kHz, the experimental result shown in Fig. 17 is obtained. In the diagram are also shown the theoretical values of the frequencies of maximum cancellation (in Hz), eqn. 42, and the frequencies of the limit of cancellation, eqn. 41, on either side of f_{mn} . As seen, the theoretical and experimental values agree closely with each other.

It is deduced from the results of the above experiment that measurement of cancellation as a function of frequency can be used, with an acceptable amount of accuracy, as a means of measuring the distance difference Δr from the sources to the observation point O and consequently can be used for measuring the difference in time τ

between the primary and secondary waves, of the same frequency, reaching the observation point. Thus it will provide a basis for measuring (unknown) separation between the primary and secondary sources.

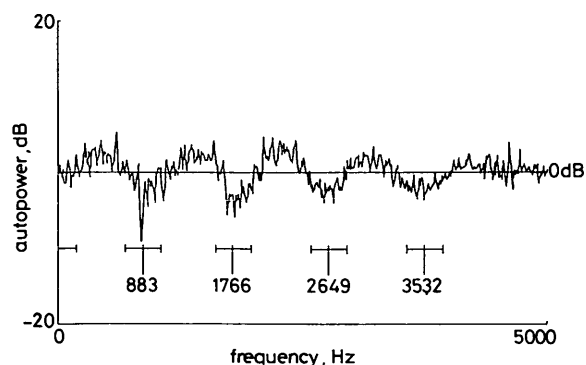


Fig. 17 Field cancellation
 $d = 1.3$ m, $\Delta r = 0.385$ m, $\alpha = -6.0$ dB

The family of cancellation curves in Fig. 10 was verified using a fixed frequency and varying the value of the power ratio for each distance difference value. The results of such an experiment using a fixed frequency of 100 Hz agree with Fig. 10 to within experimental accuracy.

4 Active noise control structures

From a practical point of view, the process of actively attenuating (or cancelling) an unwanted sound wave through wave interference consists of three main steps; detection, negation, and superposition.

In the process of detection the aim is to obtain a signal coherent with the unwanted noise over the frequency range of interest. In this case, the device to be used as a detector is required to have a response characteristic to provide this information. Microphones which have a reasonably flat amplitude characteristic and a linear phase characteristic are commonly used as detectors. Here the detector is placed at a fixed distance relative to the source of noise (primary source). The amplitude and phase of the signal will be altered owing to the acoustic properties of the acoustic path between the primary source and the detector.

The process of detection can also take place by obtaining, indirectly, a signal that is coherent with the noise [82, 86]. For example, many sources of noise vibrate continuously when in operation at a rate coherent with the acoustic waves they emit. In these circumstances, vibration-sensitive devices can be used as detectors. This method eliminates the effects of the acoustic properties of the transmission path on the unwanted noise wave and can give good results in these situations.

The most important part, and the main body of an ANC system, is the negation of the signal. The device performing this task (referred to as the controller) should be capable of not only shifting the phase of each frequency component of the detected signal by 180° but also of adjusting the amplitude of each component. The controller is, therefore, defined by a continuous transfer function representing the required amplitude and phase characteristics. Thus, when the wave of the secondary source is superimposed on that of the primary source, destructive interference results at the observation point. The required controller characteristics are dependent on the characteristics of the transmission paths from the primary and secondary sources to the detection and

observation points as well as on the characteristics of the secondary source. Therefore, the measurement of these characteristics is essential in obtaining a suitable controller.

After the detected signal has been processed by the controller, the output of the controller is used to drive the secondary source which produces an acoustic wave that interacts with the acoustic wave from the primary source. Here a loudspeaker placed at a set distance from the primary source can be used as a secondary source. The result of superimposing the acoustic waves can be observed by using an observer microphone.

From the above discussion it follows that processes of detection and superposition are relatively straightforward whereas the main task in developing an ANC system is the design of a suitable controller. Owing to the lack of a suitable design methodology and a restricted electronic technology, the design of a proper controller has been the main impediment in the success of the previous attempts at ANC.

The various arrangements of the detector, controller and secondary source have resulted in various control structures for ANC systems which essentially are classified into two basic types of control structures, namely feedback control structure and feedforward control structure.

4.1 Feedback control structure

The basic feedback control structure (FBCS) is shown schematically in Fig. 18. The primary source emits an

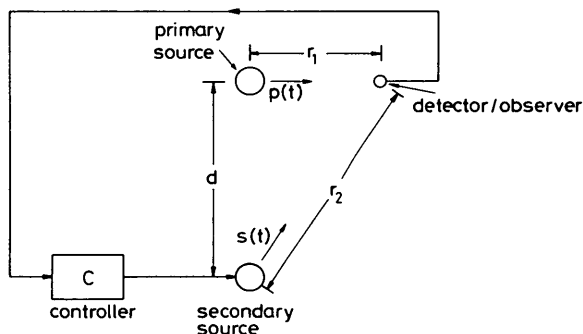


Fig. 18 Schematic diagram of feedback control structure

(unwanted) wave $p(t)$. This wave is detected by a transducer (detector), placed at a fixed distance relative to the primary source, and transferred to a controller C . The controller is required to adjust the phase as well as the amplitude of each frequency component contained in the detected signal such that when emitted by the secondary source it results in zero sound pressure level at the detector location. Hence, the detector can at the same time be the observer.

If we let

- $X_1(s)$ = transfer function of the space between primary source and detector
- $X_2(s)$ = transfer function of the space between secondary source and detector
- $M(s)$ = transfer function of detector
- $C(s)$ = transfer function of controller
- $L(s)$ = transfer function of secondary source

where s is the complex frequency in the Laplace transform, then Fig. 18 can be shown in block diagram form as in Fig. 19. This diagram shows clearly the feedback nature of the structure in that the detected signal $D(s)$ contains both the primary and secondary signals. From

this structure, the transfer function between the detected signal $D(s)$ and the primary source signal $P(s)$ is obtained as

$$\frac{D(s)}{P(s)} = \frac{X_1(s)}{1 - M(s)C(s)L(s)X_2(s)} \quad (43)$$

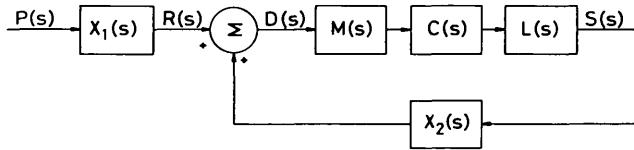


Fig. 19 Feedback control structure

Assuming the controller reverses the polarity of the signal, this can be written as

$$\frac{D(s)}{P(s)} = \frac{X_1(s)}{1 + M(s)C'(s)L(s)X_2(s)} \quad (44)$$

where $C'(s) = -C(s)$ which corresponds to a standard negative feedback structure. The objective is to reduce the detected signal to zero. Therefore, using the Schwartz inequality, it is required to have

$$|D(s)| \leq |1 + M(s)C'(s)L(s)X_2(s)|^{-1} |X_1(s)| |P(s)| \quad (45)$$

which implies that the factor

$$|1 + M(s)C'(s)L(s)X_2(s)|^{-1}$$

should tend to zero, which in turn implies that $C'(s)$ or, equivalently, the controller $C(s)$ should have infinite gain; i.e.

$$|C(s)| = \infty \quad (46)$$

In practice this is not feasible as the frequency characteristics of $M(s)$, $L(s)$, $X_2(s)$ and $C(s)$ itself will induce instability in the loop. The maximum value of $|C(s)|$ can be determined from the Nyquist stability criterion, and for satisfactory operation a suitable stability margin is required. This represents the major limitation of the feedback control structure: to achieve good cancellation a high-gain controller is required but this can produce instability in the system. Therefore, in designing the controller the loop formed by $M(s)$, $C(s)$, $L(s)$ and $X_2(s)$ should be studied from a stability point of view to ensure safe operation of the system. This structure corresponds to that proposed by Lueg in diagram 2 of his patent [8] (Fig. 1b) where the controller is realised by an electronic transmission line providing a constant time delay and, hence, providing phase inversion of a particular frequency. Similarly, Olson employed the feedback control structure in his electronic sound absorber [9] with an amplifier as a controller, providing only gain adjustment and no frequency compensation adjustment. This accounts for the limited range of operation of his device. In general, the controller is required to implement a full complex frequency compensation to achieve robust performance.

4.2 Feedforward control structure

A schematic diagram of the feedforward control structure (FFCS) is shown in Fig. 20. The primary source emits an (unwanted) wave $p(t)$. A detector placed at a distance r_1 relative to the primary source and a distance r_2 relative to the secondary source detects this signal and transfers it to the controller C . After the detected signal has been adjusted in phase and amplitude it is emitted by the sec-

ondary source to be superimposed on the unwanted noise. The aim is to reduce the sound pressure level to zero at an observation point which is at a distance of r_3

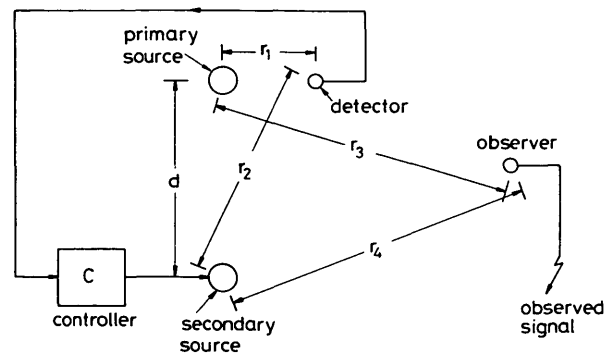


Fig. 20 Schematic diagram of feedforward control structure

relative to the primary source and a distance r_4 relative to the secondary source.

Fig. 20 is shown in block diagram form in Fig. 21 in which the transfer functions are as follows:

- $X_1(s)$ = transfer function of path r_1
- $X_2(s)$ = transfer function of path r_2
- $X_3(s)$ = transfer function of path r_3
- $X_4(s)$ = transfer function of path r_4
- $M(s)$ = transfer function of detector
- $C(s)$ = transfer function of controller
- $L(s)$ = transfer function of secondary source

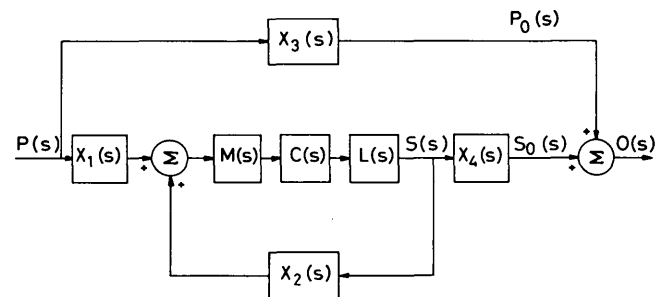


Fig. 21 Feedforward control structure

where s is the complex frequency in the Laplace transform.

In the control structure of this type, the secondary path, i.e. through $C(s)$ to the observation point, attempts to compensate in parallel with the primary path, i.e. through $X_3(s)$ to the observation point, such that as a result of the superposition of the primary and secondary waves cancellation is achieved at the observation point.

As seen in Fig. 21 there is also a closed feedback loop present in the structure which is due to the secondary wave reaching the detector. This loop can make the system unstable, as discussed previously in connection with the FBCS.

The objective with this structure is to reduce the observed signal to zero. This requires the observed primary and secondary signals to be equal in amplitude and opposite in phase; i.e.

$$P_o(s) = -S_o(s) \quad (47)$$

From Fig. 21 it follows that

$$P_o(s) = X_3(s)P(s)$$

and

$$S_o(s) = X_4(s) \frac{M(s)C(s)L(s)X_1(s)}{1 - M(s)C(s)L(s)X_2(s)} P(s)$$

Substituting the values of $P_o(s)$ and $S_o(s)$ from the above equations into eqn. 47 yields

$$X_3(s) = - \frac{X_1(s)M(s)C(s)L(s)X_4(s)}{1 - M(s)C(s)L(s)X_2(s)} \quad (48)$$

and therefore extracting an expression for $C(s)$:

$$C(s) = \frac{X_3(s)}{M(s)L(s)[X_2(s)X_3(s) - X_1(s)X_4(s)]} \quad (49)$$

This represents the desired controller transfer function for obtaining cancellation over the frequency range of interest. However, again stability conditions must be satisfied and can be verified through the use of the Nyquist stability criterion. A stability analysis of this structure has been carried out by the authors and is soon to be submitted for publication.

The feedforward control structure corresponds to diagram 4 of Lueg's patent [8] (Fig. 1d) where the time delay is implemented by the physical separation of the primary and secondary sources. Obviously this will only be accurate for a single frequency signal with proper gain adjustment. This structure has also been employed by Conover [18], Hesselman [20], Ross [21] etc. in the cancellation of transformer noise. Conover used the periodic nature of transformer noise to implement parallel delay and gain adjustments on the fundamental component and harmonics of the detected signal. Ross used a development similar to the above, however, his success was limited owing to the distributed nature of the transformer noise. He also reported a large variation in the observed cancellation due to the time-varying characteristics of the source.

Note, in Fig. 20, that if the observation point is moved to coincide with the detection point, the FFCS will be reduced to the FBCS. Therefore, from the two structures the more general one, namely the FFCS, is chosen for controller implementation, which includes the FBCS as a special case.

5 Implementation

The development of microprocessor technology provides the opportunity to develop practical ANC systems. In particular, there now exists powerful digital processors specially designed for signal processing and digital filtering applications, such as the Intel 2920 and Texas Instruments TMS32010 and TMS32020. These have the capability for the real-time processing of signals within the frequency range suitable for ANC and are cheap enough to allow dedication to one system. With such devices it is now possible to overcome many of the problems associated with previous attempts to implement ANC systems.

The Intel 2920 signal processor is a single chip microprocessor designed specially to process real-time analogue signals and has onboard program memory, digital to analogue (D/A) conversion circuitry, analogue to digital (A/D) conversion circuitry, digital processor, and input/output (I/O) circuitry. It is composed of three major sections: the EPROM, arithmetic and analogue I/O sections. The EPROM section includes an instruction clock generator and program sequence counter. The

arithmetic section, which includes a 40 word by 25 bit random access memory (RAM) with two ports and an arithmetic and logic unit (ALU), executes commands from EPROM, thereby performing digital simulation of analogue functions in real time. The analogue section, which includes a (4-input) input multiplexer, an input sample-and-hold (S&H) circuit, a D/A convertor, a comparator and an output multiplexer with 8 output S&H circuits, performs A/D and D/A conversions on commands from the EPROM section. The limits of the Intel 2920 performance are established by the size of the onchip EPROM and RAM, the speed and capability of the processor and the resolution of the A/D and D/A converters.

5.1 Digital implementation of controller

The transfer function of the controller for the FFCS (see Figs. 20 and 21), as found previously, is dependent on the transfer characteristics of the secondary source, the detector, the acoustic paths from the detector to the primary and secondary sources, and the acoustic paths from the observer to the primary and secondary sources. With $s = j\omega$ these can be represented as

$$X_1(j\omega) = H_{x1}(\omega)e^{j\theta_{x1}(\omega)} \quad (50a)$$

$$X_2(j\omega) = H_{x2}(\omega)e^{j\theta_{x2}(\omega)} \quad (50b)$$

$$X_3(j\omega) = H_{x3}(\omega)e^{j\theta_{x3}(\omega)} \quad (50c)$$

$$X_4(j\omega) = H_{x4}(\omega)e^{j\theta_{x4}(\omega)} \quad (50d)$$

From the acoustic properties of a nondispersive propagation medium it is clear that the amplitude of a signal, propagating through a distance r , is attenuated by an amount inversely proportional to the square of the distance r and is delayed in phase by an amount directly proportional to the product of the distance r and frequency ω . Thus, if the proportionality constants for amplitude and phase are, respectively, Ka and Kp , the transfer functions in eqns. 50a-d can be written in terms of r and ω . Consider the transfer functions $X_3(j\omega)$ and $X_4(j\omega)$:

$$H_{x3}(\omega) = \frac{Ka}{r_3^2} \quad H_{x4}(\omega) = \frac{Ka}{r_4^2} \quad (51)$$

and

$$\theta_{x3}(\omega) = Kp\omega r_3 \quad \theta_{x4}(\omega) = Kp\omega r_4 \quad (52)$$

If the observer location is restricted to such points that the distance difference $|r_3 - r_4|$ is much smaller than (say, less than a tenth of) the smallest of r_3 and r_4 then the transfer functions $X_3(j\omega)$ and $X_4(j\omega)$ will be approximately equal for the low-frequency range (say, 0-500 Hz) of concern in the design of ANC systems. Using this assumption, the required controller characteristic given in eqn. 49, for $s = j\omega$, can be simplified to yield

$$C(j\omega) = \frac{1}{L(j\omega)M(j\omega)[X_2(j\omega) - X_1(j\omega)]} \quad X_3(j\omega) = X_4(j\omega) \quad (53)$$

For a specific detector location relative to the primary and secondary sources, the transfer functions $LMX_1(j\omega)$ and $LMX_2(j\omega)$ can be measured over a specified frequency range and substituted in eqn. 53 to yield the required controller transfer function for that frequency range. The arrangement shown in Fig. 22 with a condenser-type microphone as a detector, at a negligible

distance from the face of the primary source (a Linn Sara loudspeaker) and about 34 cm away from the centre of the secondary source face (a Linn Sara loudspeaker having characteristic $L(j\omega)$), was used. The transfer functions $LMX_1(j\omega)$ and $LMX_2(j\omega)$ were measured, using a Solartron SP 1200 signal processor, and used to calculate $C(j\omega)$ from eqn. 53. Note that in the arrangement of Fig. 22 the loudspeakers' driver amplifier (a NAIM amplifier), the microphone amplifier 1, and the antialiasing and reconstruction filters of the Intel 2920 signal processor, represented by SDK-2920, are also included in $LMX_1(j\omega)$ and $LMX_2(j\omega)$. The inclusion of these components in $LMX_1(j\omega)$ and $LMX_2(j\omega)$ is to compensate for their effects in the performance of the ANC system. The required amplitude and phase characteristics of the controller over a frequency range of 0–500 Hz are as shown by the solid-line curve in Fig. 23. The required gain of the controller has a high value at low frequencies, resulting from the loudspeaker characteristics, and remains approximately flat above 100 Hz. The required phase characteristic, on the other hand, starts with a value well below zero degrees and steadily increases for increasing frequency. From 50 to 500 Hz the phase completes half of the 2π cycle.

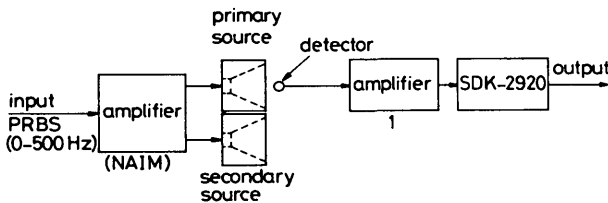


Fig. 22 Arrangement for measuring $LMX_1(j\omega)$ and $LMX_2(j\omega)$

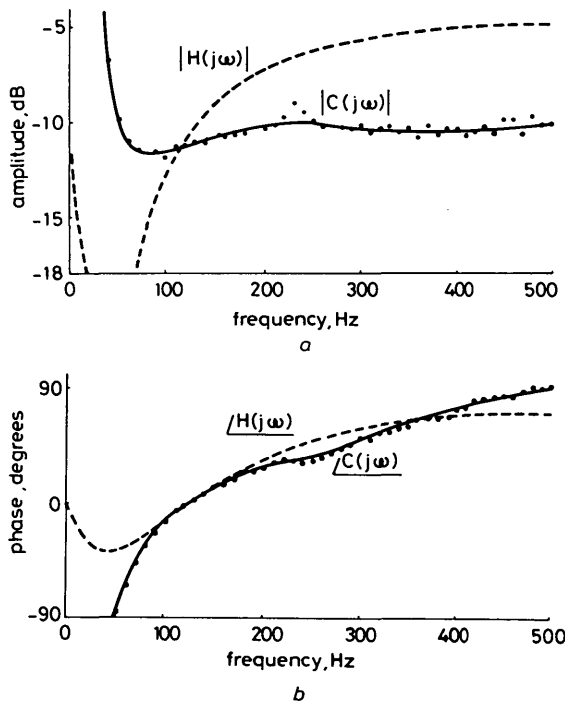


Fig. 23 Controller characteristics

a Amplitude
b Phase

Applying the condition of cancellation dictated by the range of permissible power ratio (see eqn. 19 for $\beta = 1$) to the FFCS of Fig. 21 the following is obtained:

$$0 < \left| \frac{L(j\omega)M(j\omega)X_1(j\omega)C(j\omega)}{1 - L(j\omega)M(j\omega)X_2(j\omega)C(j\omega)} \right|^2 < 4 \left(\frac{H_{x3}^2}{H_{x4}^2} \right) \quad (54)$$

substituting

$$L(j\omega)M(j\omega)X_1(j\omega) = A_1(\omega)e^{j\theta_1(\omega)}$$

$$L(j\omega)M(j\omega)X_2(j\omega) = A_2(\omega)e^{j\theta_2(\omega)}$$

$$a_x = \frac{H_{x4}}{H_{x3}}$$

and letting the offset in the controller characteristics from the required values be by a factor of $a_c(\omega)$ in the amplitude and an amount of $\phi_c(\omega)$ in the phase, then eqn. 12 becomes

$$0 < a_c^2(\omega) < \frac{A_1^2(\omega) + A_2^2(\omega)[1 - 2 \cos \phi_c(\omega)]}{0.25a_x^2 A_1^2(\omega) - A_2^2(\omega)} - \frac{2A_1(\omega)A_2(\omega)[\cos(\theta_1(\omega) - \theta_2(\omega)) - \cos(\theta_1(\omega) - \theta_2(\omega) - \phi_c(\omega))]}{0.25a_x^2 A_1^2(\omega) - A_2^2(\omega)} \quad (55)$$

which is the required relationship between the amplitude error and phase error in the controller characteristics at any frequency for the particular $LMX_1(j\omega)$ and $LMX_2(j\omega)$ transfer functions, as measured to calculate $C(j\omega)$ in eqn. 53, and the $|X_4(j\omega)|/|X_3(j\omega)|$ value.

To implement $C(s)$ as a digital filter, a suitable transfer function in the complex frequency s is required that approximates the required amplitude and phase characteristics of the controller. In choosing such a function, care must be taken to ensure the system's stability and causality; i.e. the number of zeros must be less than or equal to the number of poles. Also, the practical limitations of the Intel 2920 signal processor must be observed. After a consideration of these requirements, the fifth-order transfer function

$$H(s) = K_s \frac{(s + 298\pi)(s + 300\pi)(s + 358\pi)(s + 400\pi)(s + 462\pi)}{(s + 60\pi)(s + 100\pi)(s + 1300\pi)(s + 1400\pi)(s + 1500\pi)} \quad (56)$$

where K_s is a positive real number, was found to approximate the required controller characteristics. These characteristics are shown, for $s = j\omega$, by the broken-line curves in Fig. 23. Fig. 24 shows the offsets $a_c(\omega)$ and $\phi_c(\omega)$ in the controller characteristics from the required values. The maximum value of $\phi_c(\omega)$ is about 20° above 80 Hz whereas the amplitude error $a_c(\omega)$ can be changed by using an amplifier in the path of the controller without altering the phase characteristic, such that condition 55 is satisfied over a broad frequency range and satisfactory cancellation is obtained.

Using the bilinear z -transformation to obtain an equivalent digital filter in the recursive form [111–114] and a sampling frequency of 6.5 kHz determined by the Intel 2920, the fifth-order continuous-time transfer function $H(s)$, eqn. 56, yields its equivalent discrete-time transfer function

$$H(z) = K_z \frac{(1 - 0.8656z^{-1})(1 - 0.8648z^{-1})(1 - 0.8408z^{-1})(1 - 0.8237z^{-1})(1 - 0.7991z^{-1})}{(1 - 0.9714z^{-1})(1 - 0.9528z^{-1})(1 - 0.5219z^{-1})(1 - 0.4944z^{-1})(1 - 0.4679z^{-1})} \quad (57)$$

where z represents a complex number $|z|e^{j\omega T}$ with T as the sampling interval, and K_z is a real number. $H(z)$ was implemented as a digital filter on the Intel 2920 signal

processor using 192 instructions and setting the cutoff frequencies of the lowpass antialiasing and reconstruction filters at 16 kHz. The amplitude and phase characteristics

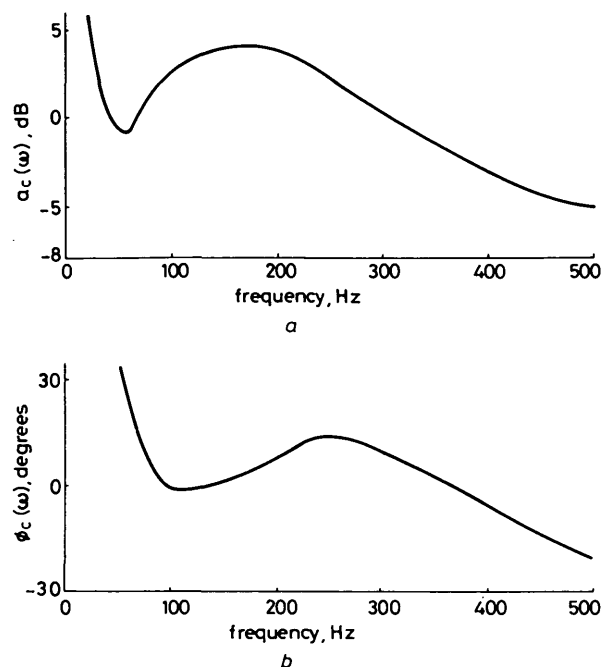


Fig. 24 Offset in controller characteristics from required values
a Amplitude error
b Phase error

of this filter, hereafter referred to as the digital filter controller (DFC), as compared to the amplitude and phase of $H(z)$, eqn. 57, are shown in Fig. 25 over a linear frequency range of 0–1000 Hz.

6 Experimental results

For the purposes of comparison, a second Intel 2920 signal processor was programmed, using 192 Intel 2920 instructions, to give a constant characteristic of unity gain and 0° phase shift over the frequency range 0–500 Hz. This is referred to as the constant-gain controller (CGC). To test the performance of the ANC system, the arrangement shown in Fig. 26 was used in which the secondary source, loudspeaker represented by $L(j\omega)$, the detector (condenser-type microphone), power amplifier and amplifier 1 are the same components as used in Fig. 22. The controller includes the Intel 2920 signal processor and the antialiasing and reconstruction filters. Amplifier 2 is used to amplify the controller output to a required level before feeding it to the driver amplifier the gain of which is controlled by the (10 k Helipot) potentiometer.

6.1 Loudspeaker as a source of noise

The performance of the DFC and the CGC was investigated using an Isobarik loudspeaker as the primary source in the ANC system of Fig. 26. The primary source was driven by a PRBS signal of bandwidth 0–500 Hz, simulating a broadband white-noise source. With the CGC the system gives the cancellation performance shown in Fig. 27a. The 0 dB line partitions the cancellation, in dB, (below the line) and reinforcement, in dB, (above the line) regions. As seen, cancellation starts at about 62 Hz and extends to about 320 Hz. The overall average cancellation provided is 3 dB with the maximum peak cancellation being 19.6 dB at 163 Hz. Here the secondary (cancelling) path of the system plays the role of a

constant-length transmission line. Thus the maximum cancellation condition is satisfied for only one frequency f_m (163 Hz). Noise components at frequencies smaller or

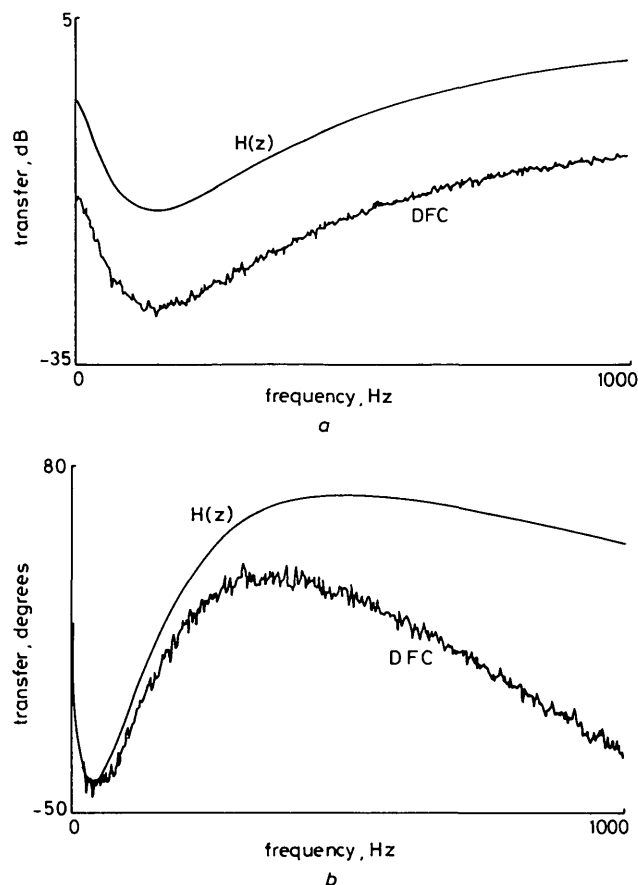


Fig. 25 Transfer characteristics of DFC compared to $H(z)$
a Amplitude
b Phase

larger than f_m are less attenuated and eventually reinforced. Ideally the curve in Fig. 27a should be symmetrical about a vertical line at $f_m = 163$ Hz. However, owing to the response characteristics of the secondary source loudspeaker, the cancellation is not symmetrical. This effect is more obvious at frequencies lower than about 60 Hz in Fig. 27a. The reinforcement at frequencies above f_m occurs owing to the linear phase shift introduced by the acoustic transmission path between the primary source and the detector and the distance difference from the observer to the primary and secondary sources. These are, of course, not considered in the design of the CGC.

Using the DFC with the transfer function given in eqn. 57, the cancellation result shown in Fig. 27b is obtained. Here cancellation starts below 50 Hz and extends to the maximum noise frequency of the source (500 Hz). Besides broadening the frequency range of cancellation from less than 2.5 octaves to more than a decade, the DFC provides a greater amount of cancellation; the average cancellation is more than 8 dB, with the maximum peak cancellation of 23.5 dB occurring at 484 Hz.

With the DFC in the ANC system, the secondary path of the system is no longer equivalent to a constant-length transmission line but has frequency-dependent characteristics that alter the amplitude as well as the phase of each detected noise frequency component such that satisfactory cancellation results are obtained. Examining the performance of the system in terms of the amplitude and phase of the controller as depicted in Fig. 23, it is seen

that by changing the gain level in the controller path the characteristic $|H(j\omega)|$ can very closely approximate $|C(j\omega)|$ such that relation 55 is satisfied over a broad fre-

in an increase in the cancellation of the primary noise for frequencies below 300 Hz. However, as Fig. 24 implies, this will deteriorate the performance of the DFC for fre-

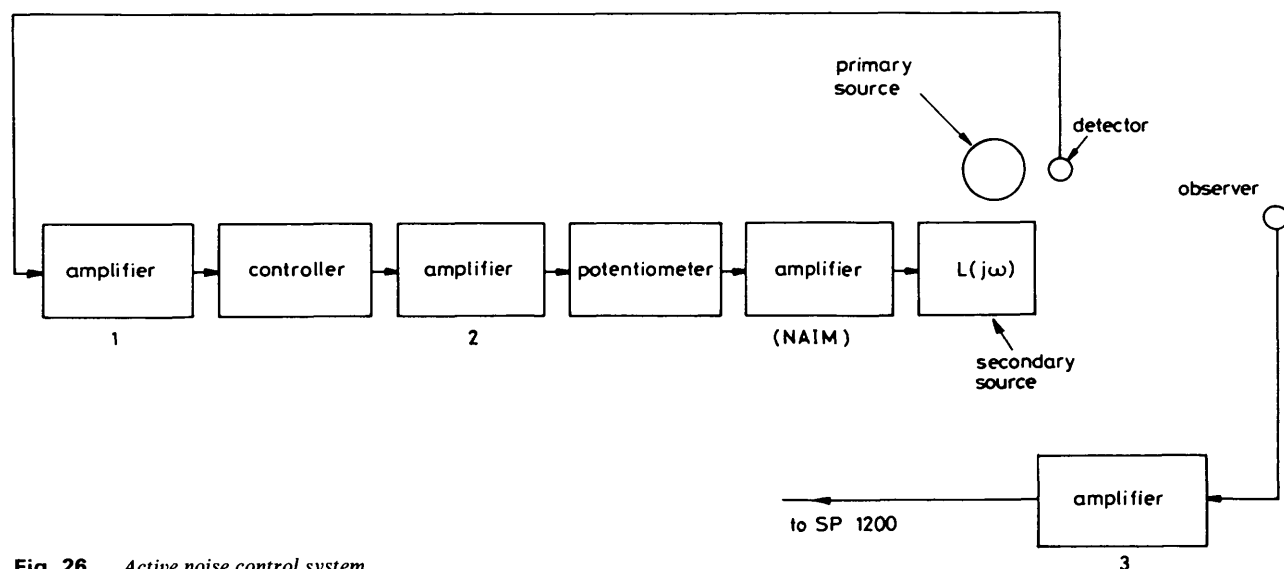


Fig. 26 Active noise control system

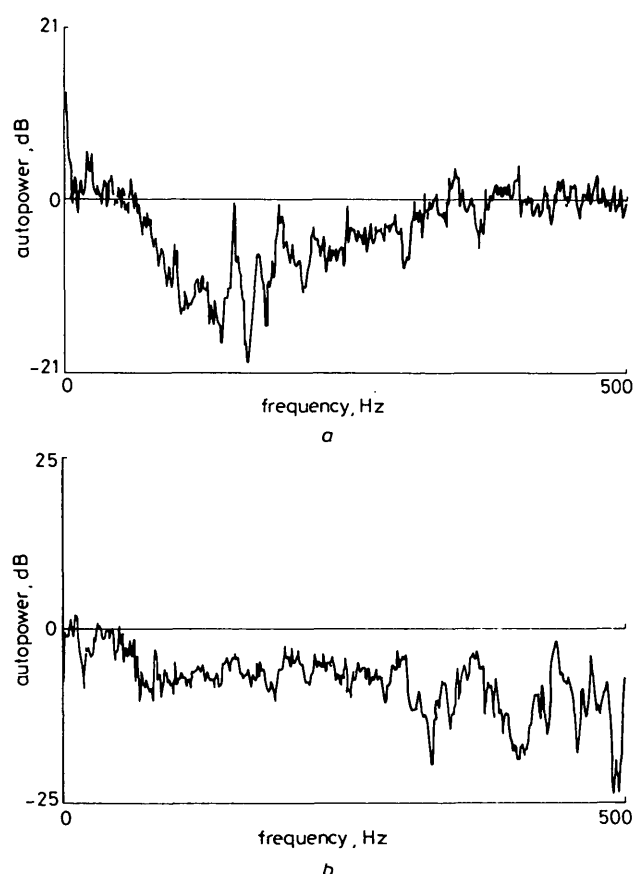


Fig. 27 Cancellation performance of ANC system with synthetic source, driven by PRBS signal

a Performance with CGC
b Performance with DFC

quency range. Owing to the relatively larger offset of $H(j\omega)$ from $C(j\omega)$ at frequencies lower than 50 Hz, no cancellation is obtained in that region. The relatively smaller amount of cancellation at frequencies less than 300 Hz in Fig. 27b is a consequence of the larger amounts of error in the amplitude $a_c(\omega)$ (which accounts for the power ratio α) in that range, as shown in Fig. 24. Altering the gain in the path of the controller can result

in an increase in the cancellation of the primary noise for frequencies greater than 300 Hz and hence worsen the overall cancellation of the primary noise.

As predicted, the performance of the DFC shows a considerable improvement over the CGC. Essentially, this is achieved by incorporating a model of the secondary source (loudspeaker) and the acoustic transmission paths in the design of the controller. The performance could be further enhanced by using the more powerful Texas Instruments TMS32010 or TMS32020 device.

6.2 Motorcycle exhaust as a source of noise

The performance of the DFC in the ANC system of Fig. 26 was investigated with a real source of noise. The exhaust noise of a motorcycle (Yamaha 100 cc), shown in Fig. 28, was chosen as this is a loud low-frequency compact source. In addition the frequency content of the noise can be adjusted by altering the throttle setting, thereby allowing the performance under time-varying conditions to be assessed. Fig. 29 shows the spectral content of the exhaust noise with the motorcycle in the idle condition. The spectrum is essentially low frequency, less than 450 Hz, and broadband, although periodic frequencies occur at intervals related to the firing frequency. Using the CGC, the performance of the system is as shown in Fig. 30a. A significant amount of cancellation occurs in the range 72–240 Hz with a maximum cancellation of 19.9 dB at 140 Hz. The reinforcement in the noise level increases with frequency above 240 Hz making the overall average cancellation slightly more than 0 dB. As compared with Fig. 27a, the performance, Fig. 30a, has deteriorated from the simulated results. This is because of a difference in the location(s) of the detector and/or observer relative to the primary and secondary sources, in measuring Fig. 30a, as compared with the situation in measuring Fig. 27a. In measuring Fig. 30a the detector and observer are located, relative to the primary and secondary sources, such that the difference in time τ_2 between the primary and secondary signals of the same frequency reaching the observer is greater, in magnitude, than the time difference τ_1 in measuring Fig. 27a. As a result of this, the frequency at which maximum cancellation occurs in Fig. 30a as well as the frequency of

the limit of cancellation (0 dB) are less than those in Fig. 27a.

Using the DFC the performance of the system, as shown in Fig. 30b, is significantly improved. Cancellation

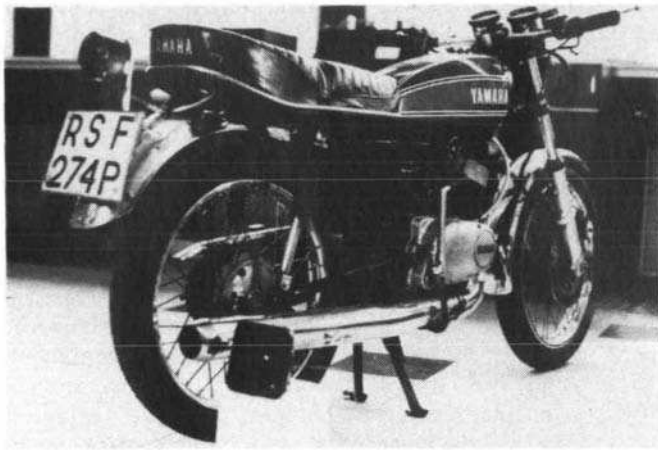


Fig. 28 Motorcycle exhaust used as noise source in ANC system

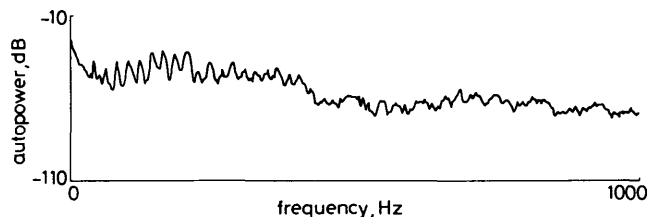


Fig. 29 Spectral density of motorcycle exhaust

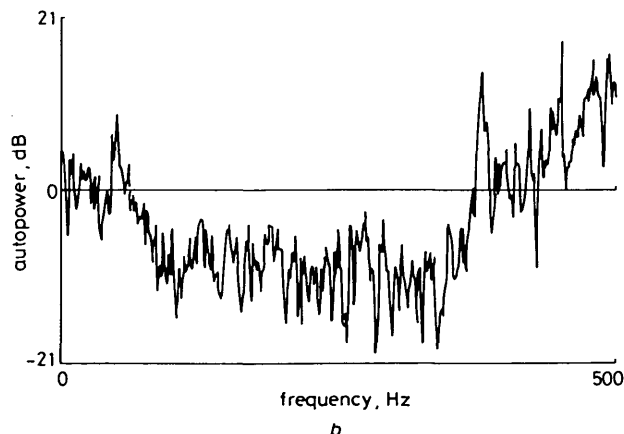
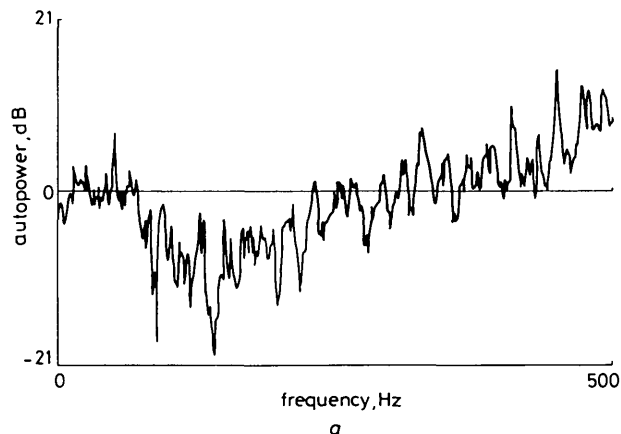


Fig. 30 Cancellation performance of ANC system with motorcycle as noise source

a Performance with CGC
b Performance with DFC

is obtained from about 62 to about 373 Hz. The overall average cancellation is more than 4 dB with a maximum peak cancellation of 19.9 dB occurring at 284 Hz.

In this experiment the DFC significantly increased the amount of cancellation over a broader frequency range. However, as compared with the simulated result, Fig. 27b, reinforcement occurred at frequencies greater than 373 Hz. This is because of the difficulty in locating the detector relative to the primary and secondary sources in the same manner as used in measuring the required controller characteristics. At very low frequencies the effect of the detector location will not be significant; i.e. the additional phase shift introduced due to the error in locating the detector is negligible at very low frequencies, but as the frequency increases this additional phase shift becomes more significant so that it eventually produces reinforcement.

7 Summary and future work

Previous development of ANC systems has tended to focus on minimising the practical problems of system implementation by developing complex geometrical structures: multiple-detector and/or multiple-loudspeaker configurations. Only limited success has been attained however. Recent advances in semiconductor technology have produced extremely fast, dedicated, digital signal processing devices capable of implementing, in real time, controllers of significant complexity. The opportunity now exists to trade geometric complexity with controller complexity, resulting in a number of significant advantages. This approach requires a proper analysis of ANC systems from a control engineering perspective, giving a clear classification of the characteristics of the sources and taking into account the acoustic response of the propagation medium and electronic equipment used in the system. Three basic characteristics of noise sources have been defined:

- (i) broadband sources
- (ii) distributed sources
- (iii) time-varying sources.

Consideration of each of these leads to a different control solution, and the general case of all three requires an extremely complex control analysis. The limited success of previous attempts at ANC can be seen as a failure to account for one or more of the above characteristics. Moreover, two other factors, namely

- (iv) acoustic response of the medium
- (v) response characteristics of transducers and other electronic equipment

have a major role in the performance of an ANC system and require careful consideration in the design stage. A signal propagating through the medium is altered in amplitude as well as in phase. This incorporated with the geometric arrangement of the system can result in poor performance and even instability in the system. A similar situation can arise owing to the response characteristics of transducers and other electronic equipment used in the system. Moreover, inside an enclosure, the multimode response of the enclosure is another factor which has an influence on the performance of an ANC system.

The unwanted noise emitted by practical noise sources is generally of broadband nature. In reducing the level of such a noise a controller, which is the main part of the system, is required to be able to produce a mirror image of every noise frequency component such that superimposition on the unwanted noise results in reduction of the

noise level. This means that the controller is required to have a proper continuous transfer function. The dependence of the characteristics of the controller on measurable frequency-dependent parameters of the system (such as the characteristics of the source, transducers and other electronic equipment, and response characteristics of the medium) makes it possible to obtain the required frequency-dependent characteristics of the controller. An ANC system suitable for the reduction of the noise of compact broadband noise sources has been designed in the form of an FFCS. This system has been tested with various noise sources. The effect of such frequency-dependent characteristics was shown to significantly increase the amount as well as the bandwidth of cancellation in each case. This fully supports the initial claim that the controller is required to have a proper frequency-dependent characteristic to improve the performance of the ANC system to a desired level.

The spectral contents of most practical sources are generally not constant but depend on external parameters, resulting in a time-varying spectrum. In reducing the noise of a compact source with time-varying characteristic, a fixed digital filter controller is not adequate. In such a case the controller is required to be able to adapt its characteristics, in accordance to the source, to maintain maximum cancellation over the required frequency range. Such a system would also have the property of self tuning the controller to the desired characteristics. Research in this area is currently being undertaken by the authors.

A feature of many practical noise sources is that they cannot be considered as compact; the physical size of the source is an appreciable proportion of the wavelength of the emitted noise. The noise emission is distributed over the source. A solution to such a case would be to use a multiple-detector and/or multiple source configuration incorporating the methodology presented as the design basis. If all the three characteristics, namely broadband, time varying and distributed, are accumulated in one source, a solution (a complex one) would be the requirement of having a controller with frequency-dependent characteristics and adaptive capability incorporated within a multiple-detector and/or multiple-source configuration. Several significant problems will have to be resolved before this general case becomes a practical proposition.

8 Conclusion

Active noise control is realised by using either one or both of the basic principles of interference and absorption. The interference principle consists of the mixing of acoustic waves resulting in constructive and destructive interference which, in turn, causes intensification and weakening of the sound field, respectively. The absorption principle, on the other hand, results by synchronising the movements of the secondary source, say a loudspeaker's diaphragm, in antiphase to the unwanted noise so that the noise energy is absorbed by the loudspeaker.

The idea of active noise control has intrigued scientists and engineers in many countries for over five decades. The early efforts at ANC, prior to the mid-1960s, were unsuccessful. This was due to a poor understanding of the problem from an acoustic and control systems point of view, as well as the limitations in technology to be able to deal with the main problems associated with the design of ANC systems.

Advances in technology which began prior to the mid-1960s, and are continuing, make the active control of noise much more feasible and attractive. These include advanced control systems technology, which includes the development of adaptive algorithms; improved understanding of the physics of acoustic systems; and the development of solid-state electronics, which makes sophisticated control realisable and inexpensive.

The main part of an ANC system which needs to be given greater consideration is a design procedure for the controller. An ANC system employing an amplifier (which is equivalent to a constant-length transmission line) as a controller can provide maximum cancellation of the unwanted noise at only one frequency, with cancellation decreasing on either side of this frequency and eventually producing reinforcement. Because of the broadband nature of many acoustic sources, the controller is required to have frequency-dependent characteristics so that it is capable of producing the mirror image of every detected frequency component of the unwanted noise. The dependence of these characteristics on a number of frequency-dependent factors within the system makes it possible to measure and realise the required characteristics as a controller. An ANC system employing such a controller provides large amounts of cancellation over the required broad frequency range. In the past, because of the complexity involved in implementing such a controller, geometrical complexity, by using a number of secondary sources, was preferred. However, with the capabilities of current electronic technology, especially digital electronics, it is possible to implement such a controller either as a fixed digital controller or as an adaptive digital controller, whichever is required by the application, on a special-purpose microprocessor.

The capability of today's electronic technology provides the opportunity to develop sophisticated active noise control systems that can 'self tune' to the required control characteristics, thereby minimising the need for extensive analysis. Such systems will also allow the system to 'track' the variation in spectral content of real sources such that stable cancellation is maintained over a range of operating conditions. This adaptive capability is essential to the realisation of successful active noise control systems for real industrial applications. These techniques provide an opportunity for the electrical engineering community to contribute to the improvement of our living environment.

9 Acknowledgment

Mr. M.O. Tokhi acknowledges the support of a British Council Research Scholarship.

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